



Northwestern University

Department of Electrical and Computer Engineering

Computer Engineering 392: VLSI Systems Design Projects

Learned-Feature BMI ASIC: A Mixed-Signal Chip for Real-Time Neural Intent Classification

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Note to grader:

- Since all the Cadence design files are about 1 GB total, the paths on the Wilkinson servers, which the project is entirely designed on, are provided instead.
- The submission onto Canvas includes the digital RTL, simulation, and synthesis files (essentially everything on the digital side except for the place-and-route)
- **Path to design for all design files: /home/vcs5888/CE392/bmi_chip**
- **Path to analog design folder: /home/vcs5888/CE392/bmi_chip/analog**
- **Path to digital design folder: /home/vcs5888/CE392/bmi_chip/digital**
- The digital portion is also published on GitHub as a public repository at [this link](#), which has a README included as well
 - **https://github.com/zachtey/bmi_chip_digital**
- As this is the first time that we as a group are submitting an ASIC design on this particular server, please let us know if the design cannot be opened.
- Our final presentation can be found [here](#)
 - https://docs.google.com/presentation/d/1DCwt0UbiEcIfI-s5YcL_vL9OcoDMgItz/edit?usp=sharing&ouid=111346477464904392175&rtpof=true&sd=true

Abstract

This report presents the design of a fully integrated brain-machine interface (BMI) ASIC for real-time neural signal acquisition and motor intent classification, implemented in a 45 nm GPDK process. The chip targets intracortical neural recording applications, interfacing directly with Blackrock Neurotech's Neuroport electrode array to acquire signals from eight cortical channels simultaneously. The analog frontend consists of eight low-noise amplifiers (LNAs) based on the Harrison and Charles capacitively-coupled neural amplifier topology. The analog amplified signal is then multiplexed and converted into a digital signal into a post-processing backend. The digital backend processes the digitized samples through a pipelined datapath a Spiking Band Pass (SBP) feature extraction algorithm and is then classified by a fixed-point multilayer perceptron (MLP) with 8-bit weights, achieving 70% classification accuracy across four grasp categories derived from the Brochier macaque motor cortex dataset. The results are transmitted to an external controller over SPI. The digital block was synthesized to a 50 MHz target clock, with a critical path of 13.606 ns providing 6.13 ns of timing margin, and verified through 40-vector RTL and gate-level simulation with 100% functional accuracy. The full classification pipeline completes in under 500 μ s, supporting over 2,000 classifications per second. The end result is a compact and fully integratable brain-machine interface solution that can be integrated into implantable ASIC applications as a SPI slave or peripheral.

Introduction and Motivation

Over 5 million Americans live with paralysis resulting from stroke, ALS, spinal cord injury, or other neurological conditions. Studies from the National ALS Registry shows that estimated prevalence rates of ALS in the United States are increasing year on year, as shown in Figure 1 below. For many of these patients, voluntary motor control is permanently lost, yet the neural intent to move remains intact within the motor cortex. These neural intent messages are sent as microvolt fluctuating neural waves whose pattern can be classified to a specific neural intent.

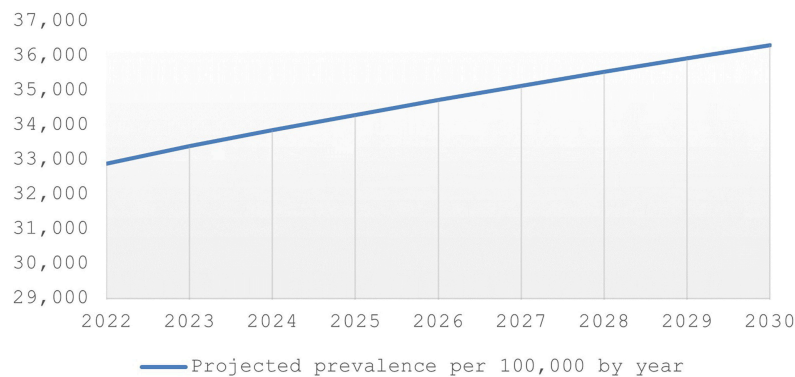


Figure 1: ALS Prevalence by Year from 2022 to 2030

Brain-machine interfaces (BMIs) offer a path toward restoring communication and motor function by reading that intent directly from neural signals and translating it into control commands for external devices such as robotic limbs, exoskeletons, or communication aids. However, realizing a clinically viable BMI requires solving a difficult hardware problem: neural signals recorded at the electrode are extremely weak, on the order of microvolts, and must be amplified, filtered, and digitized before any classification can take place. Doing this at low power, in a small form factor, and with sufficient channel count for reliable decoding demands a fully custom mixed-signal ASIC rather than off-the-shelf components.

This project presents the design of a mixed-signal BMI chip implemented in a 45 nm GPDK CMOS process. The chip reads intracortical neural signals from eight electrode channels simultaneously, using a custom analog frontend to amplify and digitize the signals, and a digital backend to extract features and classify the user's intended movement in real time.

The classification target is a four-class reach-to-grasp task derived from recordings collected by Thomas Brochier at MIT in 2018, from massively parallel intracortical recordings in macaque motor cortexes during an instructed delayed reach-to-grasp behavioral task. The four output classes are:

- PG-LF (Power Grasp, Low Force): A whole-hand grasping motion applied with low grip force, typical of picking up a lightweight or fragile object.
- PG-HF (Power Grasp, High Force): The same whole-hand grip applied with greater force, as when gripping a heavy or resistive object.

- SG-LF (Skilled Grasp, Low Force): A precision finger grasp, such as a pinch, applied gently, used for tasks requiring fine motor dexterity at low effort.
- SG-HF (Skilled Grasp, High Force): A precision finger grasp applied with higher force, for tasks requiring both dexterity and grip strength.

Distinguishing between these four classes is clinically meaningful. In neuroprosthetic applications, the difference between a power grasp and a skilled grasp determines how a robotic hand shapes its fingers, while force level determines how firmly it closes. A BMI that can decode both grasp type and force intent simultaneously allows for far more natural and functional prosthetic control than a binary open/close command. For patients with high-level spinal cord injury or limb loss, this level of dexterity could meaningfully restore the ability to perform activities of daily living.

The neural basis for classification comes from the population coding properties of motor cortex neurons. Individual neurons in areas M1, PMd, and PMv have preferred directions, meaning that they fire maximally when the intended movement aligns with their tuning direction and minimally when opposed, following approximately a cosine tuning curve. With eight electrodes placed near neurons with diverse preferred directions, the population firing pattern forms a unique signature for each intended movement class. The chip's role is to estimate each neuron's firing rate from the recorded signals and classify the resulting population pattern.

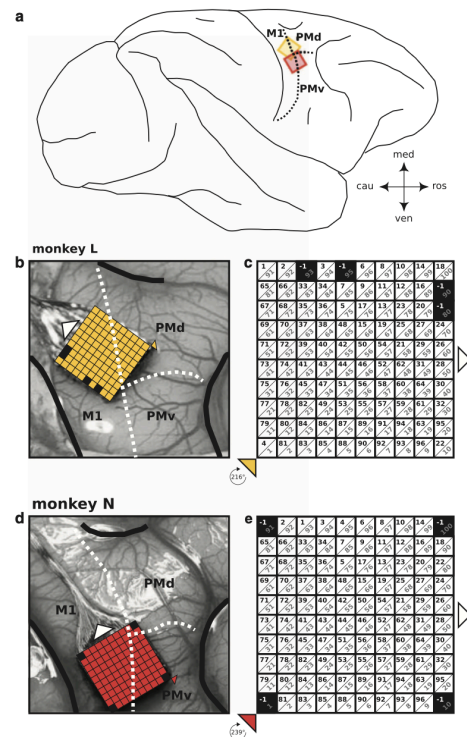


Figure 2: Brochier's Experiments with Utah Array Electrodes

A key architectural advantage of this chip is that the classifier itself is not hardwired to these four grasp classes. The MLP weights are stored in a 108-byte scan chain register file that is loaded serially at power-up before inference begins. This means the chip can be reprogrammed to classify an entirely

different set of four movement categories simply by shifting in a new set of weights with no hardware changes. A model trained on wrist rotation, finger individuation, speech articulation, or any other four-class neural decoding task could be deployed on the same silicon by loading the corresponding quantized weight file onto the scan chain. This makes the architecture broadly applicable across BMI paradigms and patient populations, and positions the chip as a general-purpose neural classification engine rather than a task-specific device.

Broader Considerations and Literature Review

The challenge of recording and decoding neural signals in a low-power, integrated form factor has driven decades of research in neural ASIC design. This section reviews the key prior works that motivate the design choices made in this chip, and identifies the gaps that this work addresses.

Harrison & Charles (2003)

The seminal work by Harrison and Charles established the capacitively-coupled neural amplifier topology that remains the dominant architecture for intracortical recording frontends. Their design, implemented in a 1.5 μm CMOS process, achieved 40 dB of midband gain, a bandwidth from 0.025 Hz to 7.2 kHz, and an input-referred noise of 2.2 μV_{rms} at 80 μW , with a noise efficiency factor (NEF) of 4.0 approaching the theoretical minimum of 2.9. The use of MOS-bipolar pseudoresistor elements elegantly solved the problem of DC offset rejection without requiring off-chip coupling capacitors, and the PMOS input pair operating in subthreshold achieved the high g_m/I_D ratio needed for low-noise operation at microamp bias currents. Thus, this has become the industry standard topology and has informed the development for many neural ASICs, including the state of the art chips in production by Neuralink and BlackRock Neurotech today. In our work, we draw upon the same circuit topology.

However, there exists drawbacks of this design as it was designed over two decades ago. This design was implemented in a 1.5 μm process, which is no longer industry standard today as process nodes have shrunk down, and consumed 0.16 mm^2 per channel, making direct scaling to high channel-count implants challenging. The authors noted that the area was dominated by passive capacitors, and that MUX and ADC design for large-scale systems remained an open problem. Our design directly addresses this by adapting the same topology to a 45 nm process, reducing area significantly while maintaining the core noise-power tradeoff, and pairing it with a fully integrated digital backend on the same chip.

Kim et al. (2021)

Kim et al. extended the Harrison-Charles topology to a 16-channel time-division multiplexed system in 180 nm CMOS, achieving 3.44 μW and 0.012 mm^2 per channel with an IRN of 4.27 μV_{rms} . Their key innovations included an area-aware LNA design methodology that accounts for noise multiplication from parasitic input capacitance, an OTA with local positive feedback to decouple G_m from noise bandwidth, and early introduction of a second pole to eliminate the need for a per-channel low-pass filter. The topology will be described in detail in the next section.

These represent meaningful advances in area and power efficiency over the baseline Harrison topology. However, this work stops at signal digitization as it outputs raw digitized neural data over SPI without any on-chip feature extraction or classification. Every sample must be transmitted to an external processor for neural decoding, which imposes a substantial wireless bandwidth and power burden on any implantable system. For a fully implantable BMI, this off-chip data pipeline is a fundamental bottleneck. Our chip closes this gap by integrating the full signal chain from electrode to classification output on a single device.

An et al. (2022)

The NeuRAD from the University of Michigan represents the closest inspiration to this work. Implemented in 180 nm CMOS, it combined a dedicated spiking band power (SBP) extraction hardware unit with an ARM Cortex-M0 processor and a fixed-point matrix acceleration unit for Kalman filter decoding, all validated in vivo with a non-human primate performing a finger control task. The system consumed just 581 μW for the full end-to-end pipeline across 93 channels, demonstrating that an implantable power budget for a complete BMI is achievable.

However, NeuRAD relied on external Intan RHD2132 analog frontend chips for amplification and digitization, consuming an additional 12 mW from three off-chip AFE chips. This means the complete recording-to-decoding system drew roughly 12.6 mW, with the analog frontend dominating power consumption at over 95% of the total. Additionally, NeuRAD used a linear Kalman filter decoder, which, while computationally efficient, has limited capacity to capture nonlinear structure in neural population activity. The SBP hardware unit also fixed the feature extraction algorithm in silicon, offering no flexibility to retarget the classifier. Our design addresses all three of these limitations: the analog frontend is fully integrated on-chip in 45 nm CMOS eliminating the off-chip AFE power overhead, the classifier is a nonlinear MLP rather than a linear filter, and the network weights are reconfigurable via scan chain allowing the same silicon to be reprogrammed for different tasks.

Bulfer et al. (2025)

The Caltech FENet work pushed the state of the art in feature extraction quality, demonstrating that a learned 1D convolutional neural network operating on broadband neural data outperforms SBP by 18% in decoding accuracy and maintains 28% better performance four years post-implant when electrode SNR has degraded significantly. Implemented in 65 nm CMOS across 192 channels, the system validated online closed-loop cursor control with a human participant implanted for six years. This work makes a compelling case that learned features substantially outperform simple bandpower metrics for long-term implant stability.

The FENet system, however, is purely a feature extraction engine with no integrated analog frontend and no on-chip decoder, as it still requires off-chip digitization and a downstream decoding system. At 1.8 μW per channel for feature extraction alone, it also requires a separate power budget for the AFE and the decoder. In addition, the architecture uses on-chip SRAM for weight storage, adding complexity relative to the scan-chain approach used here. Our chip takes a middle ground: the MLP decoder is more informative than a linear Kalman filter while being far simpler than a full 1D CNN, keeping the weight storage to 108 bytes that fit entirely in scan-chain flip-flops with no SRAM required.

Positioning of This Work

Taken together, the prior literature reveals a gap where designs with high-quality analog frontends lack integrated digital intelligence, while designs with sophisticated on-chip decoding rely on external analog frontends that dominate system power. No prior work has demonstrated a single-chip solution integrating

a fully custom analog neural recording frontend with on-chip nonlinear classification in a standard CMOS process at the channel count and power level relevant to implantable BMIs.

This challenge is the gap that this chip aims to fill. The Harrison-Charles LNA topology is adapted to 45 nm, the NeuRAD's insight of co-locating feature extraction and decoding on one device is extended to include the analog frontend, and the algorithmic approach draws from both the SBP literature and NeuRAD's demonstrated efficacy of fixed-point MLP classifiers for neural intent decoding.

Broader Considerations

As we look at the potential use cases of this chip, it is beneficial to understand the external factors that would inform the ethical application and feasibility of doing so.

The clinical BCI market is growing rapidly, driven by early commercial devices from companies such as Synchron, Neuralink, and Blackrock Neurotech. A fully integrated, low-power mixed-signal neural classification ASIC directly addresses the power and form factor constraints that currently limit these systems. The reconfigurable classifier architecture, where any four-class neural decoding task can be targeted by loading a new weight file, further broadens the addressable market beyond a single clinical application.

The impact of this technology is potentially life changing for some. For patients with high-level spinal cord injury, ALS, or locked-in syndrome, a fully implantable device capable of decoding motor intent in real time could restore meaningful independence, control of a computer cursor, a robotic arm, or a communication device. Reducing the system to a single chip lowers the barrier for miniaturization and eventually for fully wireless, battery-powered implants that do not require a transcutaneous connector, which is a major infection risk in current clinical systems.

However, intracortical neural data is among the most sensitive personal information imaginable, encoding thoughts and intentions before they are acted upon. This raises significant concerns around data security, patient consent, and algorithmic bias. The SPI output interface on this chip transmits unencrypted data, which presents a vulnerability if the device were to be used in a clinical context. Any path to human use would require FDA medical device regulatory review, likely as a Class III device, and would need to address biocompatibility of the implant, long-term safety of the recording electrodes, and robust data encryption. Additionally, the classifier was trained on macaque data from a specific behavioral task; generalization to human patients across diverse populations and neural recording conditions would require extensive revalidation.

As our team is limited in knowledge and experience, as well as time and scope of the duration of the course (CE392) to 10 weeks, we are unable to address many of these standards, but we hope that this work will be able to serve as a proof of concept that serves as an inspiration for real work later on.

Design Considerations & Requirements

The design of this chip was governed by a set of constraints spanning signal fidelity, power, area, timing, and process technology. These constraints shaped every major architectural decision in both the analog and digital domains, as well as the interactions between the two. This section details what these challenges were, the solutions and options considered, the intuitions behind the design process, and the resulting architectural topological choices that were made as a result.

Input Neural Signals

Intracortical neural signals present a demanding acquisition environment. Extracellular action potentials have amplitudes on the order of tens to hundreds of microvolts and contain energy primarily in the 300-10,000 Hz band, while low-frequency local field potentials can extend below 1 Hz with amplitudes up to several millivolts. Large DC offsets of 1-2 V are common at the electrode-tissue interface due to electrochemical effects, and must be rejected without blocking the signal of interest. These characteristics imposed three hard requirements on the analog frontend: sufficient voltage gain to bring the signal above the noise floor of the ADC, a bandpass response covering 300-10,000 Hz, and passive DC offset rejection without off-chip components. Thus, our design was designed for a 1V DC offset, as reported by the Harrison and Charles paper.

The chip was designed to interface directly with Blackrock Neurotech's Neuroport electrode array, which defines the electrode impedance seen by the LNA input. This impedance set the input loading constraints for the amplifier and informed the pseudoresistor sizing for the low-frequency cutoff. Further theory on the circuit level application will be discussed in the next section.

Input neural signal solutions are addressed in the circuit-level implementation of the LNA, touched upon in the Design Description sections under the Analog Mixed Signal sections.

Sampling and Frequency Budget

The specific analog signals of interest also impose restrictions on the sampling, which directly impact the architecture choices of the chip. The Nyquist criterion requires a minimum sampling rate of twice the highest signal frequency of interest. With the LNA passband extending to 10,000 Hz, each channel requires at least 20 kHz of sampling bandwidth. The chip uses a single SAR ADC shared across all eight channels via a time-multiplexed analog MUX, so the ADC must operate eight times faster than the per-channel Nyquist rate. This sets a minimum ADC clock rate of 160 kHz, and the design targets 200 kHz to provide margin, yielding an effective per-channel sampling rate of 25 kHz. At 25 kHz per channel, 250 samples are collected per channel within a 10 ms window, giving a classification rate of approximately 100 Hz.

The ADC was pre-designed for 8-bit resolution with a 1 V_{pp} input range, so the amplification of the analog signal must span 0-1V to leverage the entire resolution of the ADC. The 8-bit word width also propagates into the digital backend, where all feature computation and MLP weights are represented in 8-bit fixed-point arithmetic, avoiding the need for wider data paths.

Power Constraints

The long-term viability of an implantable BMI is fundamentally constrained by power dissipation. Chronic implants must not generate sufficient heat to cause tissue damage; a heat flux exceeding approximately 80 mW/cm² can cause necrosis in neural tissue, placing a hard ceiling on total chip power for a given die area.

Power constraints are touched upon in the Design Description sections under both the Digital and Analog Mixed Signal sections.

Area and Integration Constraints

All analog and digital blocks were implemented in the 45 nm GPDK process. A key constraint was the elimination of off-chip components as there are no external coupling capacitors, no off-chip resistors, and no external memory. All MLP weights, 108 bytes total, are stored in on-chip flip-flop registers loaded via a scan chain at power-up. The smaller model size allows for less memory on chip to go towards storing the weights. Since we want to make the chip's model adaptable to new applications, we can accept the scan chain loading overhead as a one-time cost at startup, which was a conscious design choice.

The time-multiplexed architecture, in which a single ADC serves all eight channels through an analog MUX, was a deliberate area and power tradeoff. A parallel architecture with one ADC per channel would reduce latency and simplify timing but would multiply the analog circuit area and power by eight. For the target application where 10 ms windows at 20 Hz classification rates are acceptable, sequential multiplexing is well justified. The ADC design was also drastically over-specified, as it can reach up to 5 MS/s, which is over the 200 KS/s minimum for double Nyquist frequency, even with all 8 channels in consideration. Since the ADC was pre-designed (in another course as part of CE493 at Northwestern, designed by Zach), we were able to more efficiently move the design constraints onto the topology of the chip instead of changing the core parameters of the ADC.

Digital Timing Constraints

The digital backend operates across three asynchronous clock domains, each with independent constraints defined in the SDC file. As detailed above, the target clock can be slowed significantly (down to even 200 KHz) to reduce power consumption. However, during the design process, significant margins were given to ensure proper functioning as the main system clock targets 50 MHz synthesis, with a critical path of 13.606 ns leaving 6.13 ns of positive slack, a 30.7% timing margin. At the actual intended operating frequency of 10 MHz, this margin expands to over 86 ns, making the digital logic massively over-designed for its operating conditions.

This was an intentional choice to ensure that the synthesis tool at our disposal, Cadence Genus, could select reasonably optimized cells without pushing for maximum speed at the cost of area and power. The SPI clock runs at 10 MHz asynchronous to the system clock, and the scan chain clock runs at 25 MHz asynchronous to both. All three clock domains are declared as asynchronous groups in the SDC

constraints, and no timing paths are analyzed across clock domain boundaries. Input delays were budgeted at 5 ns relative to the system clock, consuming 25% of the 20 ns period, with matching output delay budgets on the ADC channel select and SPI MISO outputs.

ASIC vs FPGA

In general, our team chose to use an ASIC design instead of an FPGA for a variety of reasons. Since the application case would be more directly related to an implantable chip, rather than constantly connected to an FPGA board. The most obvious reason is that discrete analog amplification on the PCB level would take significant amounts of space, and an integrated CMOS implementation is a more noise-resilient and compact solution that exists since it removes many of the environmental factors present in discrete circuit design. In an application where the location of the design and signal integrity (along with dealing with microvolt range inputs), integrated design proved to be a more ideal solution. An upside of using CMOS is that both analog and digital portions can be combined with ease. Moreover, since the actual digital logic on the chip is relatively small (our design simplifies many designs who have over 100 channels down to an 8 channel design), we are able to reduce the total area of our chip to about 0.524 mm^2 . It also benefits the chip that since neural signals are by nature lower in frequency (peaking at 10K Hz), the chip can sample much slower and thus reduces the maximum frequency that the chip must run on, reducing power, a key factor in low-power ASIC designs, since power and frequency scales directly.

Design Description

This portion of the report details the in-depth details of the system-level choices, block level implementation choices, and verification results. This chip implements a complete end-to-end neural signal acquisition and classification pipeline on a single mixed-signal ASIC, from raw intracortical electrode voltages to a classified motor intent output transmitted over SPI. The full signal chain is divided into two major domains: an analog mixed-signal frontend responsible for amplification, multiplexing, and digitization, and a digital backend responsible for sample buffering, feature extraction, inference, and communication. Both domains are implemented in the 45 nm GPDK CMOS process. The analog frontend designed in Cadence Virtuoso and the digital backend synthesized through Cadence Genus, testbenched in Cadence Xcelium, and placed and routed with Cadence Innovus.

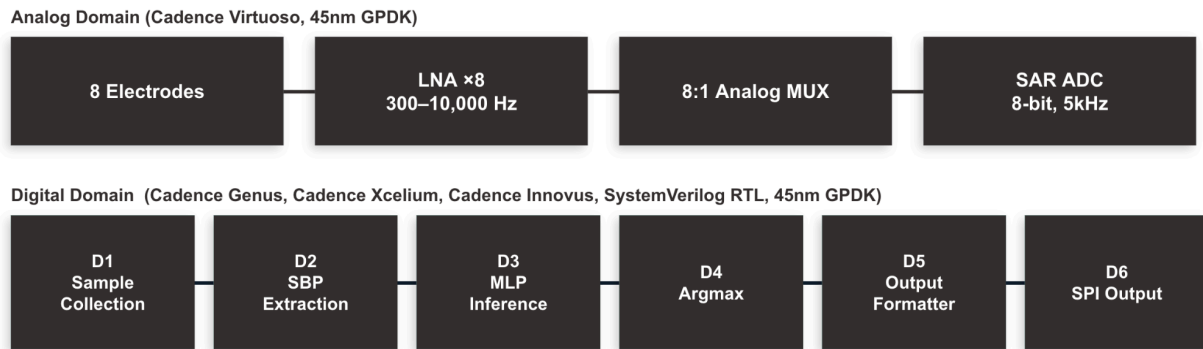


Figure 3: High Level Architecture of BMI ASIC

Figure 3 shows the high level architecture, and each sub-block will be detailed in the following sections.

Analog and Mixed Signal Design

Topology Overview

The analog front end conditions the raw intracortical signals from the eight-channel electrode array before they are digitized. Each channel uses a low-noise amplifier (LNA) chain to amplify the microvolt-scale neural signal across the 300 Hz-10 kHz band, after which a decoder-based 8:1 analog multiplexer time-interleaves the eight channels into a single 8-bit SAR ADC with a 1 V_{pp} input range. The entire analog domain was designed and simulated in Cadence Virtuoso using the GPDK045 process development kit.

The amplifier topology is a capacitively-coupled neural amplifier based on the architecture of Harrison and Charles (IEEE JSSC, 2003), which remains an industry-standard design for neural recording ASICs. Closed-loop gain is set by the ratio of the input and feedback capacitors, $C_1 / C_2 = 100$, targeting 40 dB of mid-band gain per stage. The core gain element is a current-mirror operational transconductance amplifier (OTA) with a PMOS input pair biased in subthreshold for low-noise, low-power operation. DC offset rejection is handled by a MOS-bipolar pseudoresistor chain operating in subthreshold, which presents a

compact teraohm-scale feedback impedance. This helps to block electrode DC offsets while passing the neural band, as DC offsets typically occur in the range of 1-2 V, while the signals themselves are in the range of 1-500 μ V. The topology was adapted from the original legacy micron-range process to 45 nm and sized around the electrode metallization impedance of the Neuroport array.

Operational Transconductance Amplifier (OTA)

The amplifier core is a current-mirror OTA with a subthreshold PMOS input pair. The reference Harrison and Charles design used a 1.5 μm process at a higher supply; porting to GPDK045 at a reduced single-rail supply substantially cut the available voltage headroom, so the cascode devices from the original topology were removed to recover headroom, as is visible in Figure 4.

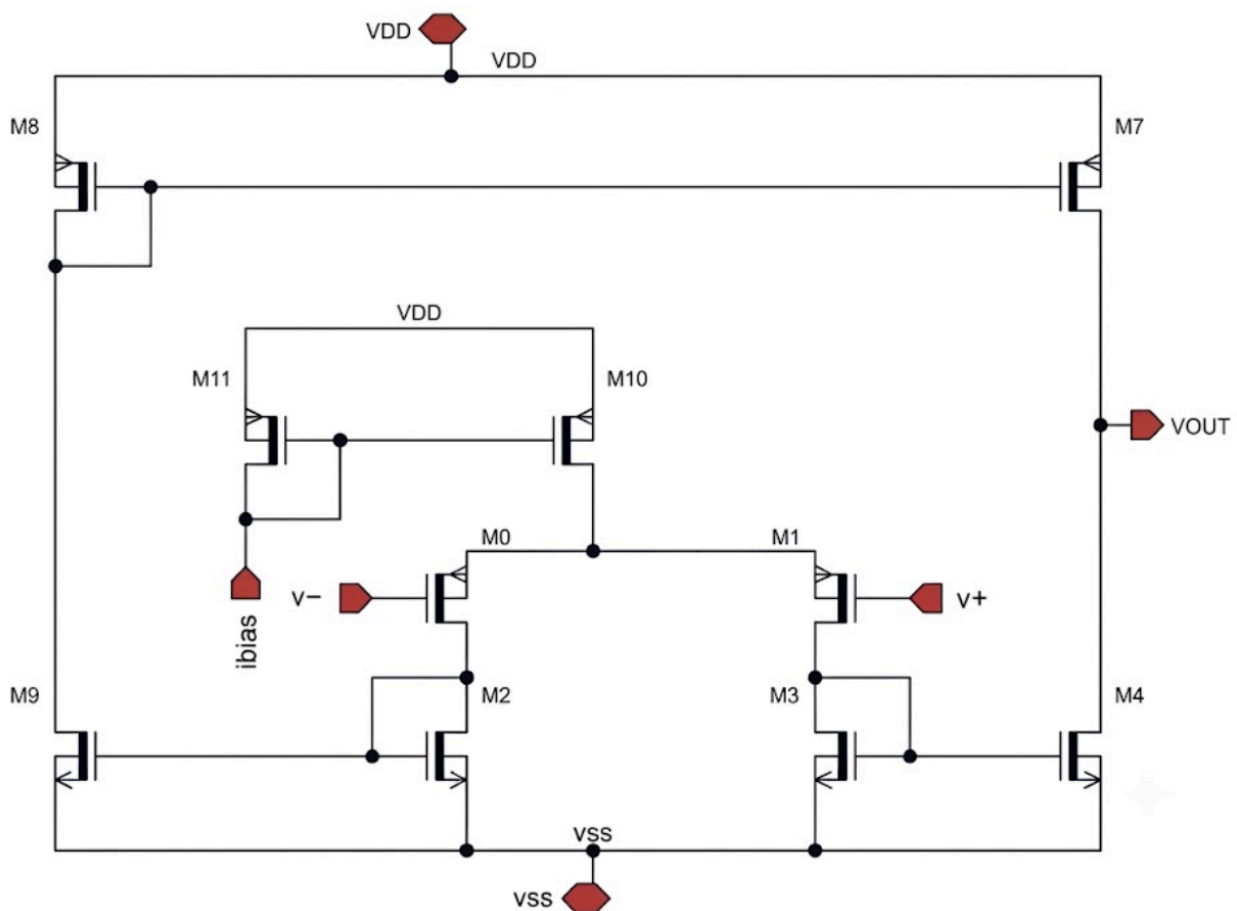


Figure 4: OTA Schematic Diagram

Since all current mirrors maintain a 1:1 ratio, we derive the transfer function by the following, using small signal analysis:

$$i_{out} = i_{d0} - i_{d1} = g_{m0}(v_+ - v_-) = g_{m0} v_{in} \Rightarrow G_m = g_{m0}$$

$$\omega_{p1} = \frac{1}{(r_{o7} \parallel r_{o4})C_L}$$

For a standard single-pole system, the typical transfer function is given as follows:

$$H(s) = \frac{A_{DC}}{1 + \frac{s}{\omega_{p1}}} = \frac{g_{m0}(r_{o7} \parallel r_{o4})}{1 + s(r_{o7} \parallel r_{o4})C_L}$$

Utilizing control theory concepts, we derive the following table:

Design Metrics	Relationship
Slew Rate	$SR = \frac{I_{bias}}{C_L}$
DC Gain	$A_{DC} = g_{m0}(r_{o7} \parallel r_{o4})$
GBW	$GBW = \frac{g_{m0}}{2\pi C_L}$
Dominant Pole	$\omega_{p1} = \frac{1}{(r_{o7} \parallel r_{o4})C_L}$
Non-Dominant Pole	$\omega_{p2} = \frac{g_{m8}}{C_{m8} + C_{m9}}$
Unity-Gain Frequency	$\omega_u = \frac{g_{m0}}{C_L}$
60° Phase Margin Condition	$\omega_{p2} \geq \sqrt{3}\omega_u$
Phase Margin Expression	$PM = 90^\circ - \tan^{-1}\left(\frac{\omega_u}{\omega_{p2}}\right)$

Table 5: OTA Physical Parameters

We also performed gm/Id sweeps across multiple lengths of N and P-channel devices to accurately size transistors, as can be seen in Figure 5.

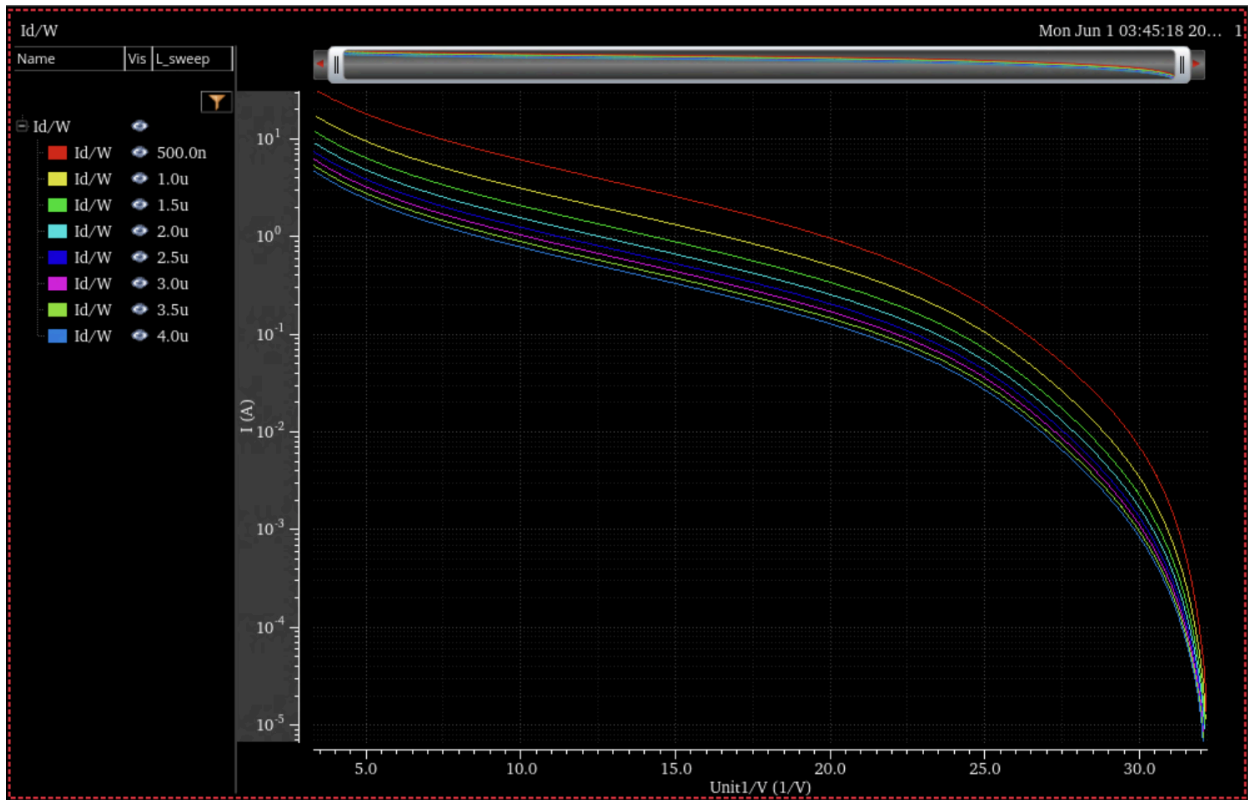


Figure 6: Sample g_m/I_d Simulation Sweep

Through the utilization of the parameters above, we implement the OTA as shown in Figure 6.

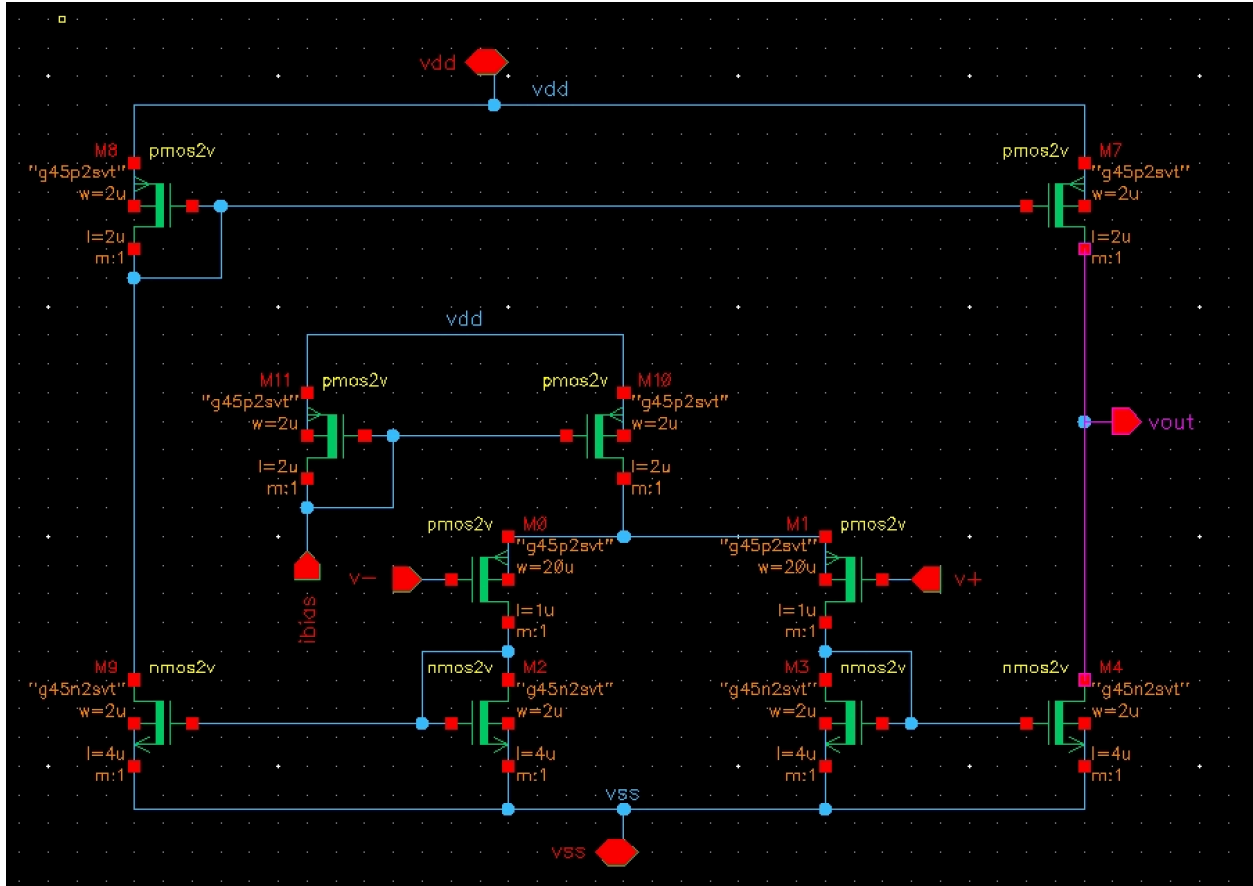


Figure 7: OTA Schematic View

The loss of output impedance from removing the cascode transistors was compensated by using long-channel current mirrors ($L = 2\text{--}4\ \mu\text{m}$), which restores output resistance and preserves open-loop gain. With this adaptation, the OTA achieves a 47 dB open-loop gain, providing sufficient loop gain for the capacitively-coupled feedback network to set an accurate closed-loop response.

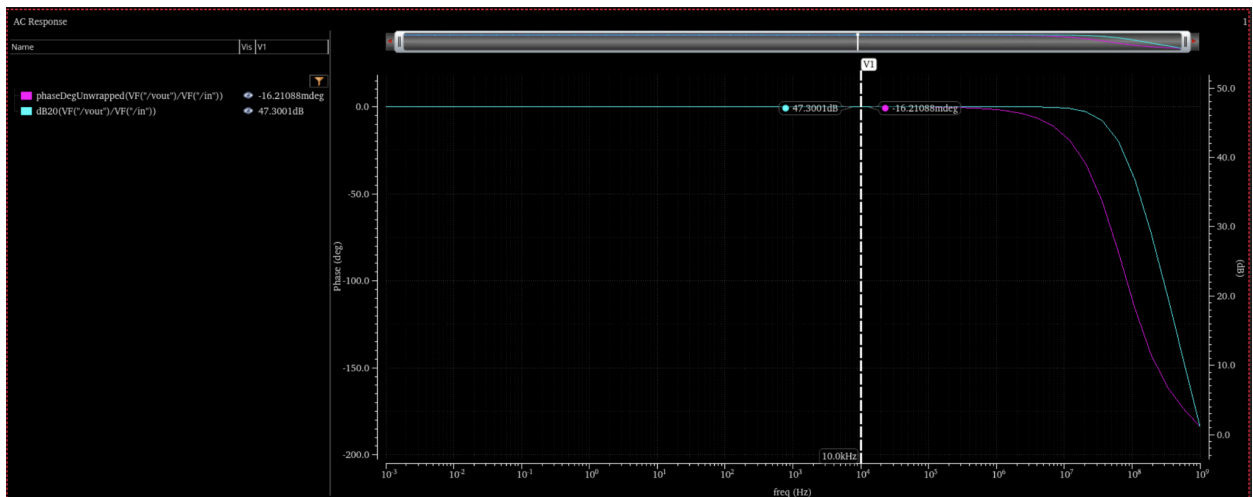


Figure 8: OTA Frequency Response

The OTA layout was built around device matching and isolation, since the amplifier's input-referred offset, noise, and gain accuracy all depend on the precise matching of the input pair and current-mirror devices. The transistors are placed in an interdigitated common-centroid arrangement with dummy devices on the array edges, as is visible in the floorplan design in Figure 7. The common-centroid geometry averages out linear process and temperature gradients across the die, so that systematic mismatch between matched pairs is largely cancelled, while the dummy transistors help to minimize edge effects due to the lithography process.

Isolation was handled with star-grounded guard rails surrounding the sensitive analog devices. Routing all guard-ring taps back to a single ground reference avoids ground loops and shunts substrate noise away from the input pair, reducing crosstalk and parasitic coupling from neighboring switching circuitry. Finally, signal routing was kept off the active channels wherever possible: minimizing metal routing over the transistor gates reduces gate capacitance and coupling onto high-impedance nodes, both of which would otherwise degrade bandwidth and noise.

Device	W	L	Type	Finger W	Fingers
M0	20	1	PMOS	2.5	8
M1	20	1	PMOS	2.5	8
M2	2	4	NMOS	1	2
M3	2	4	NMOS	1	2
M4	2	4	NMOS	1	2
M7	2	2	PMOS	1	2
M8	2	2	PMOS	1	2
M9	2	4	NMOS	1	2
M10	10	2	PMOS	2.5	4
M11	10	2	PMOS	2.5	4

Figure 8: OTA Transistor Sizing Table

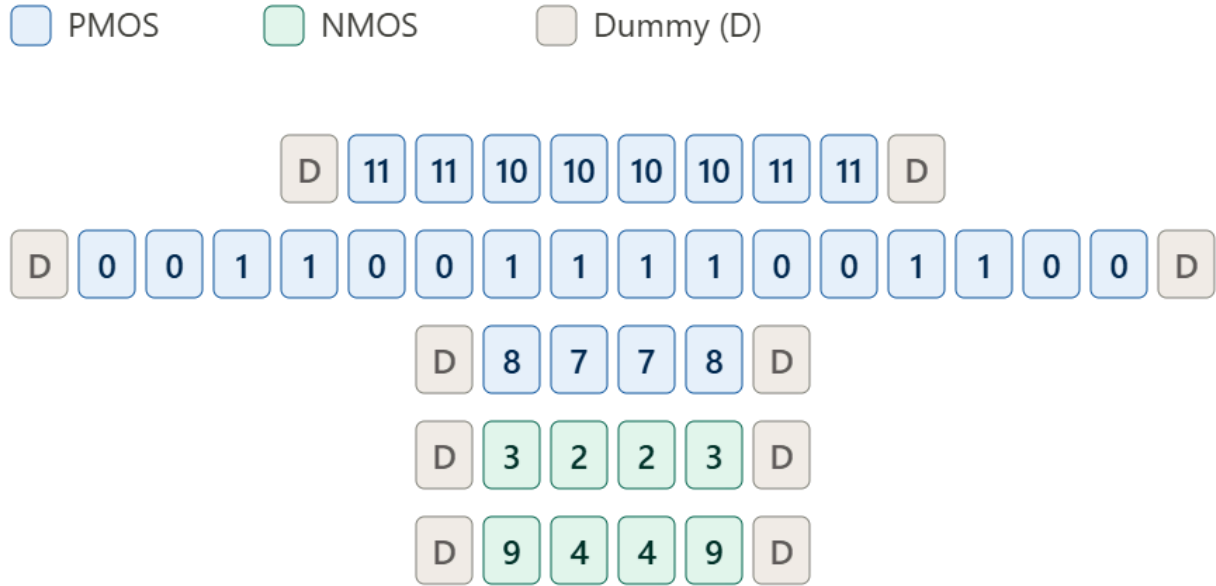


Figure 9: OTA Floorplan Design

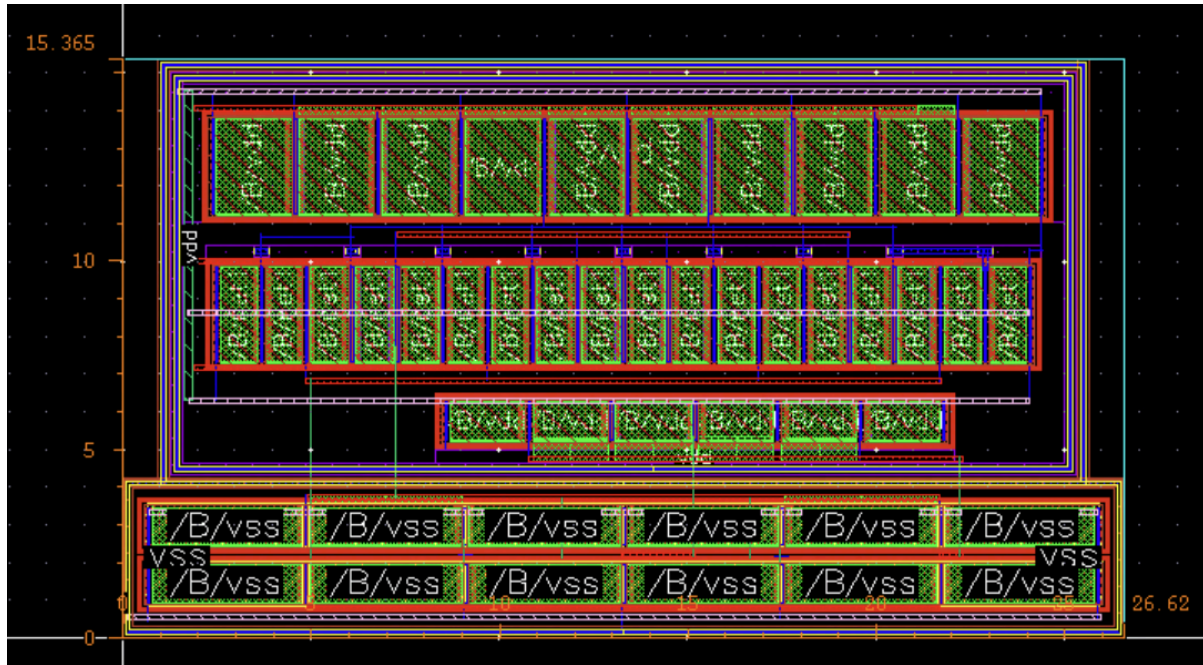


Figure 10: OTA Layout View

Low-Noise Amplifier (LNA)

The LNA wraps the OTA in the capacitively-coupled feedback network described above. It is AC-coupled, with pseudoresistors sized to operate in the deep subthreshold region so that the feedback network rejects the large intracortical DC offsets present at the electrode interface. A single stage achieves approximately 37 dB of gain with a large unity-gain bandwidth (Figure 11), reliably amplifying the neural

signal across the recording band with minimal attenuation. The high-pass corner set by the pseudoresistor-capacitor feedback blocks DC while passing the band of interest.

Input impedance was validated by testing based off of Blackrock Neurotech's Neuroport Electrode array spec sheet, which is described as having a 50 k Ω maximum value for SIROF electrodes at a signal frequency of 1 kHz. Simulating the design with these values, signal attenuation is below 1%, which we deemed acceptable for this application.

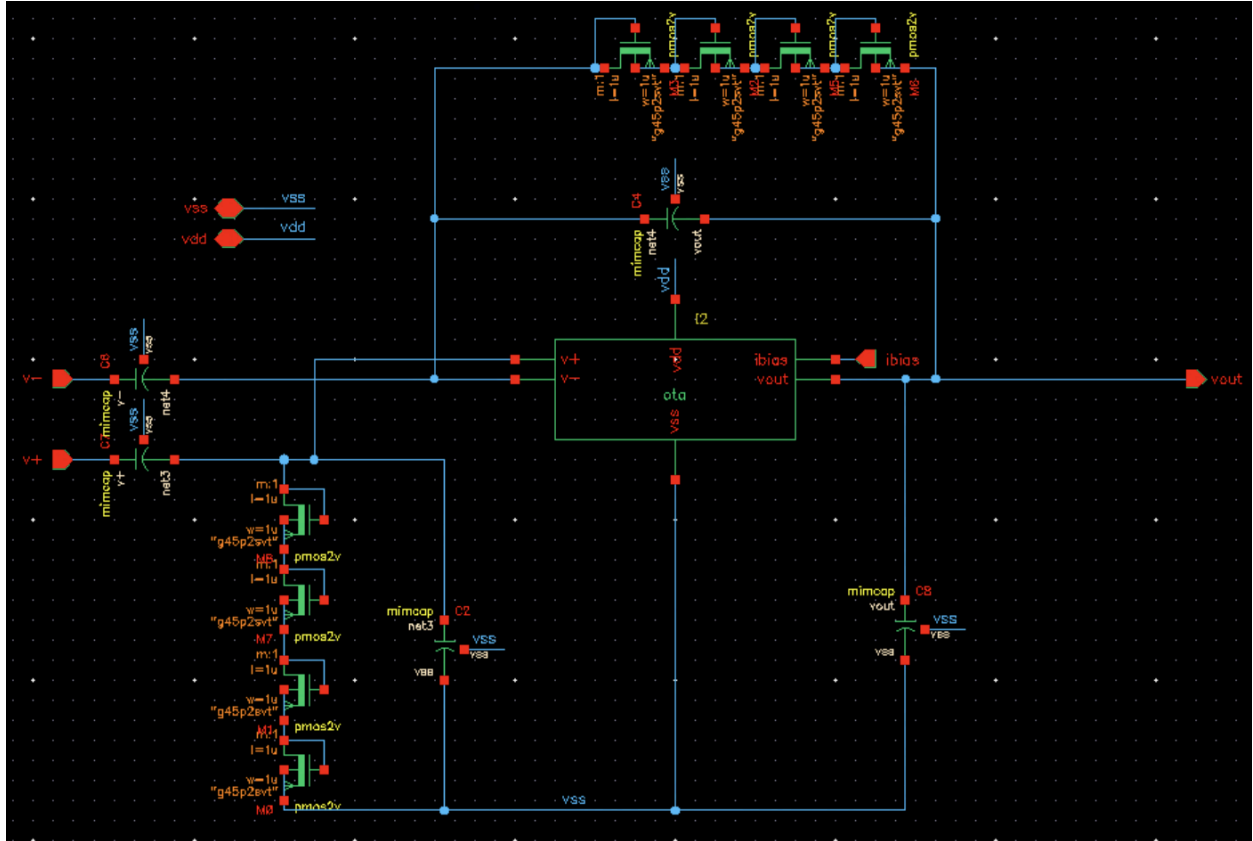


Figure 11: LNA Schematic View

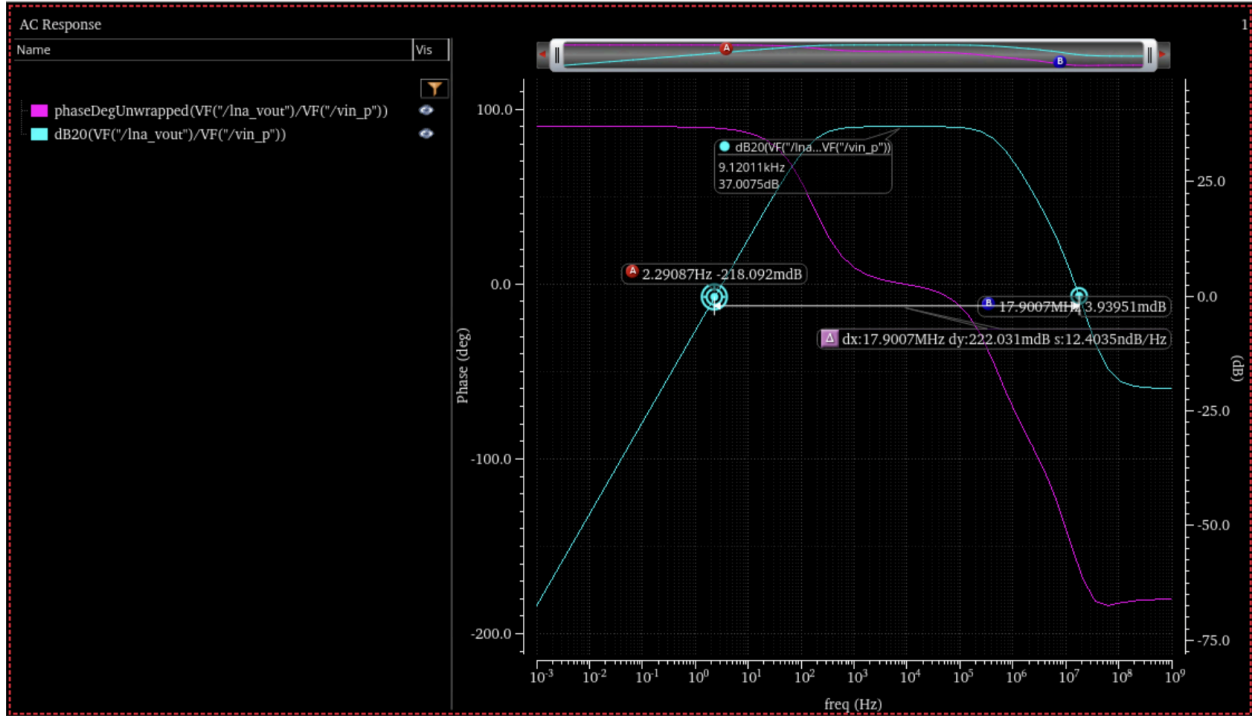


Figure 12: LNA Frequency Response and Bandwidth

The LNA layout places the matched OTA core within its surrounding capacitively-coupled feedback network: the input capacitors, the feedback capacitors, and the MOS-bipolar pseudoresistor chain. The pseudoresistor devices are placed compactly within the feedback path; their teraohm-scale impedance means they carry negligible current and occupy little area, so the passive capacitor banks dominate the LNA footprint. The resulting stage was kept dense and regular so that it could be abutted and replicated in the multichannel front end. The majority of input pins are accessible from the left of the block, while outputs are accessible from the right, such that the design is easily cascable.

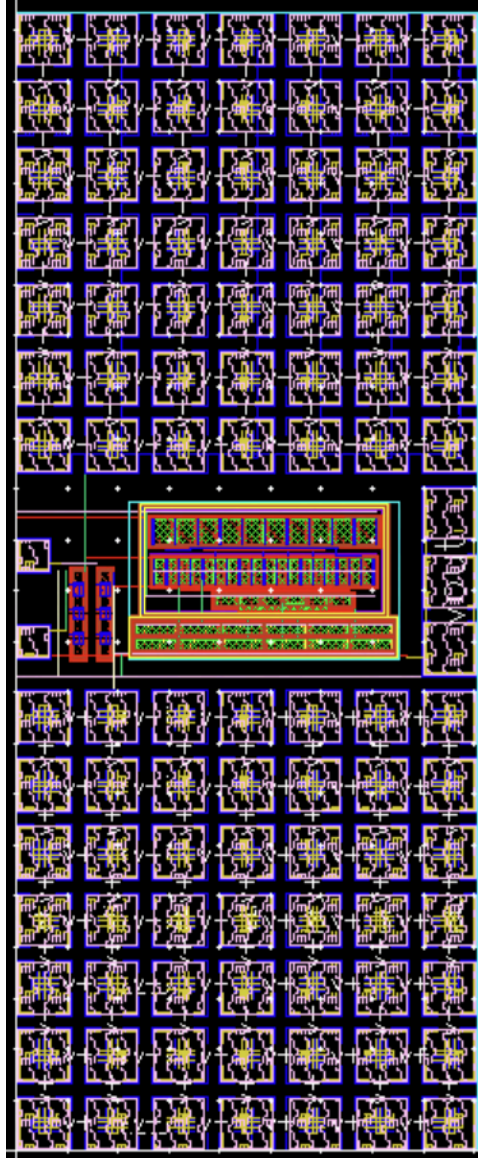


Figure 13: LNA Layout View

Cascaded LNA

A single capacitively-coupled stage provides only ~ 37 dB of gain, which is insufficient to drive the neural signal across the full ADC input range. The complete per-channel amplifier therefore cascades two LNA stages together with a DC-offset adjustment high-pass filter, which re-centers the signal between stages so that the second stage's output makes full use of the ADC's 1 V_{pp} range. Post-parasitic-extraction simulation of the cascaded amplifier shows approximately 65 dB of total gain, corresponding to over $1500\times$ amplification of the input signal, as can be visualized along with the large bandwidth in Figure 16. The stage layout is compact and tiles cleanly, making it straightforward to replicate the amplifier across all eight channels in the multichannel front end.

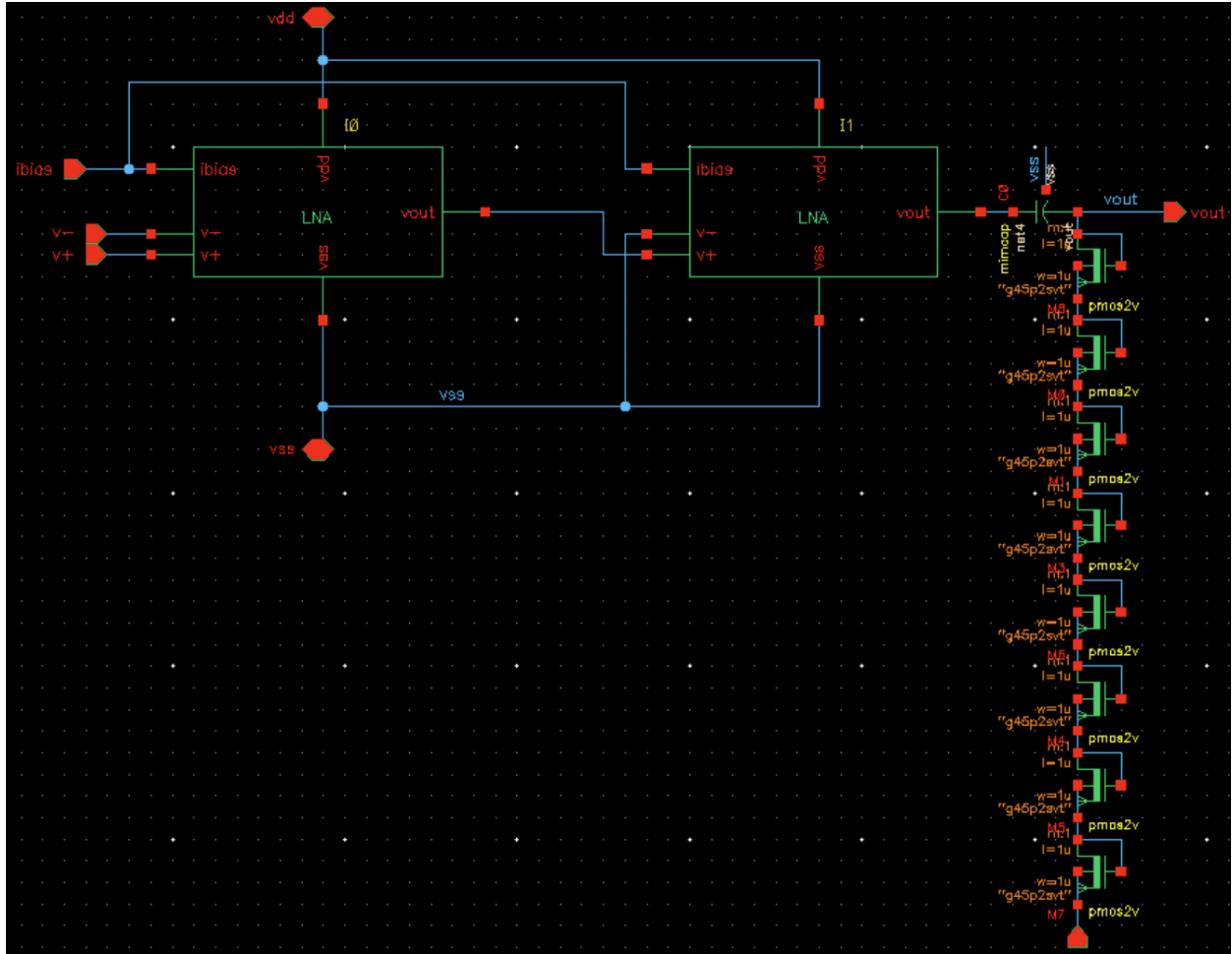


Figure 14: Full Cascaded LNA Schematic View

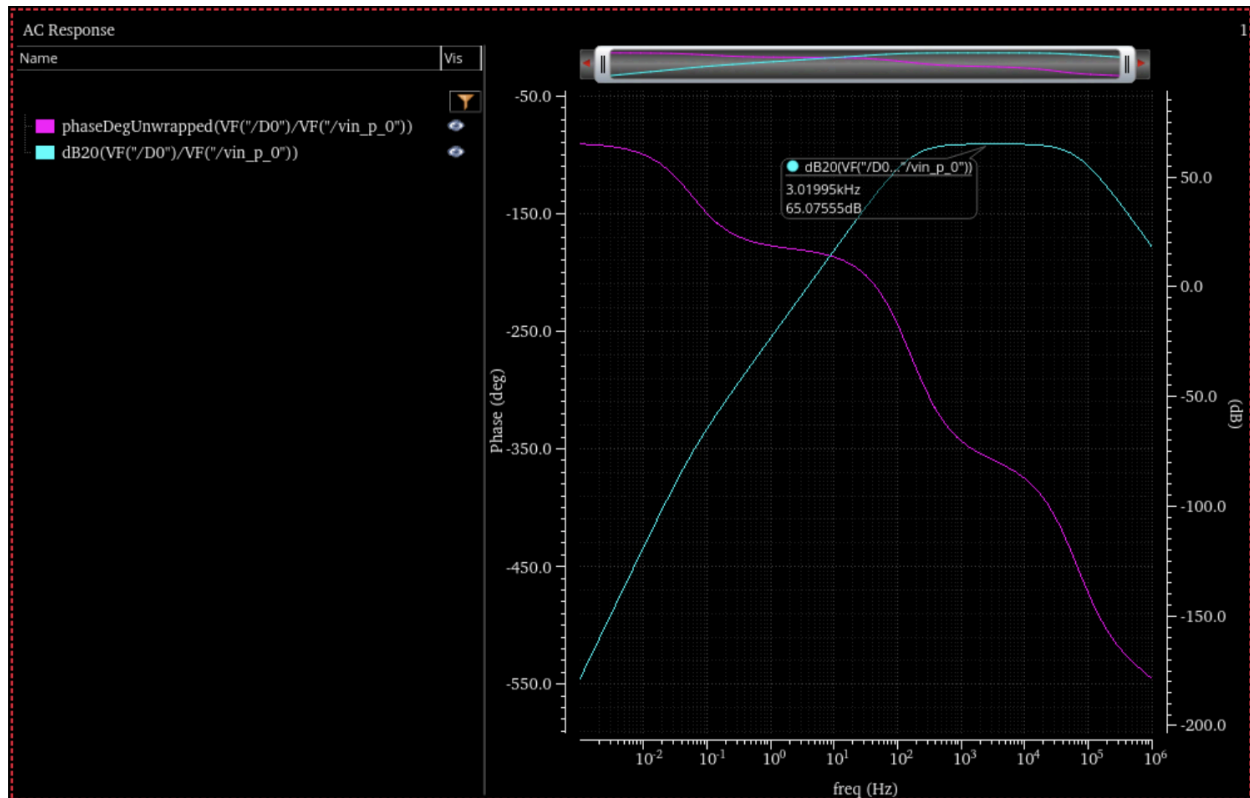


Figure 15: Full Cascaded LNA Frequency Analysis

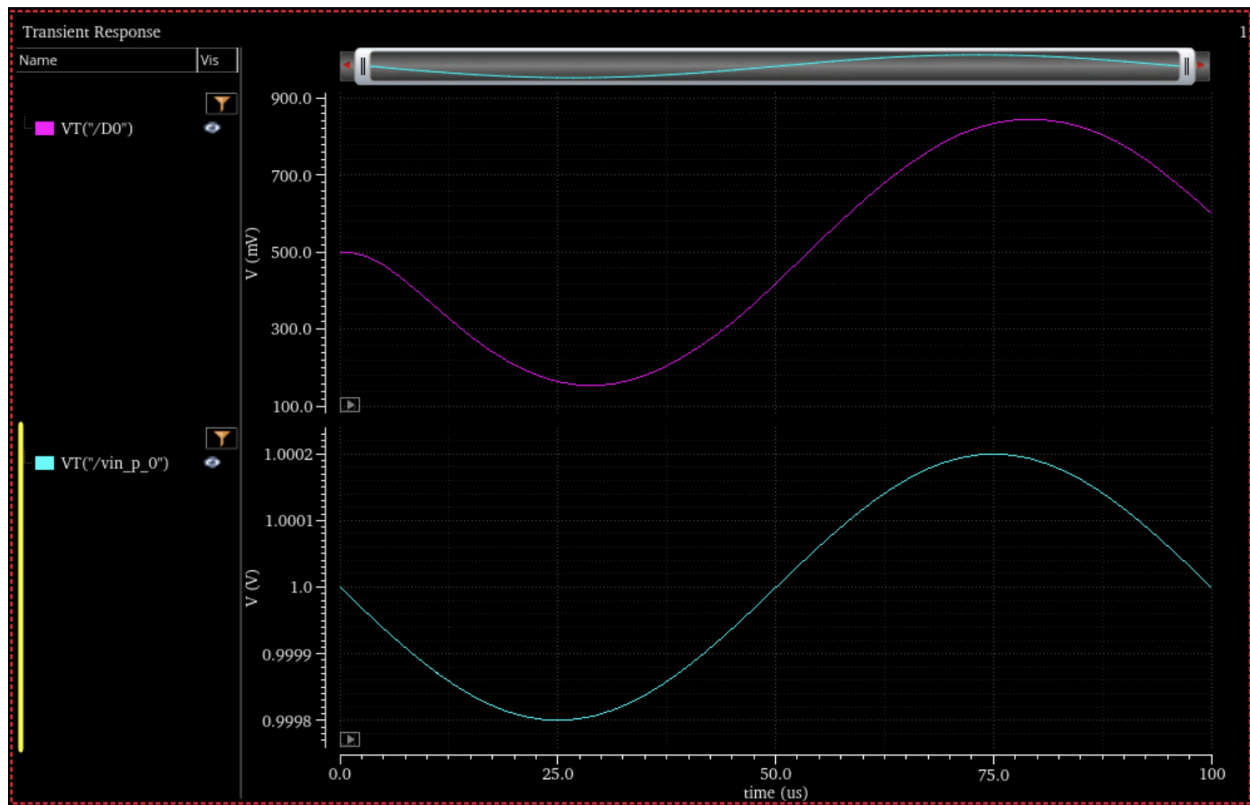


Figure 16: Full Cascaded LNA Transient Response

The complete per-channel amplifier is laid out by cascading two LNA stage layouts together with the inter-stage DC-offset adjustment high-pass filter. The two stages are placed side by side and abutted so that the inter-stage connection and shared bias and supply rails route cleanly between them, keeping the combined two-stage block compact.

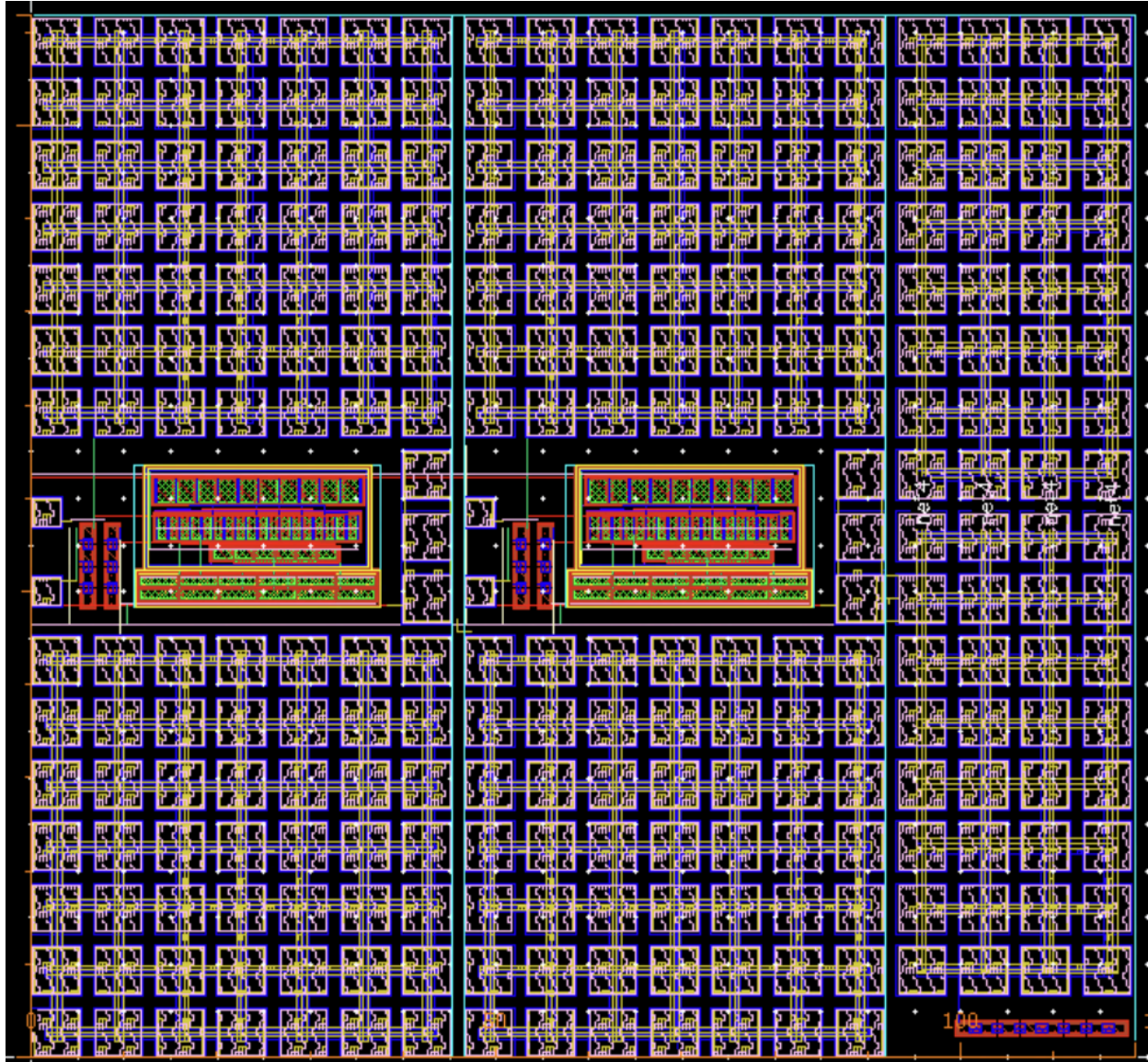


Figure 17: Full Cascaded LNA Layout View

8:1 Analog Multiplexer (MUX)

Following the amplification stage, an 8:1 Analog MUX time interleaves the neural signals from the eight LNA channels into a single shared 8 bit SAR ADC. Because the signal chain relies on a single MUX to switch between channels rapidly without degrading the signal integrity.

The core architecture of this block consists of an 8:1 digital decoder that uses 3 select pins (SEL0 - SEL2). As shown in Figure 18, the decoder feeds directly into a single transmission gate phase for each channel.

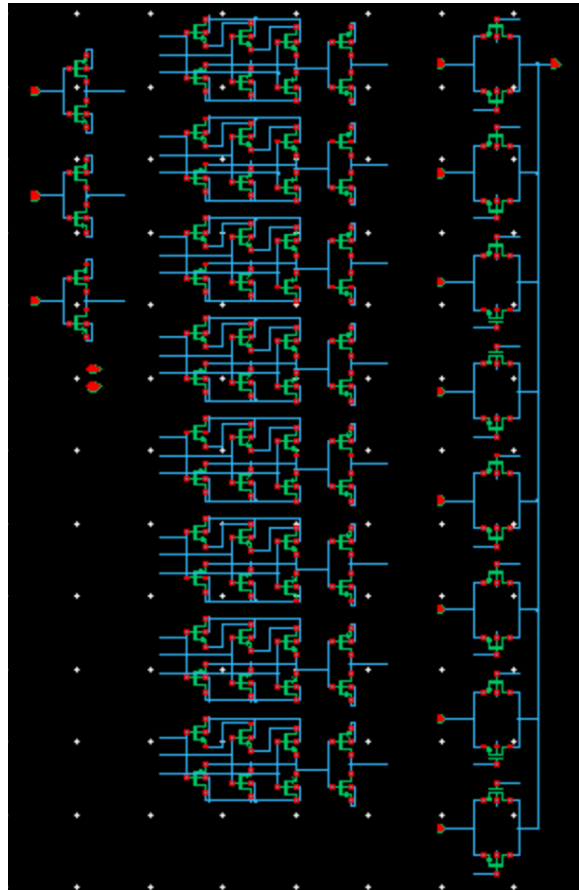


Figure 18: MUX Schematic View

In earlier designs, we tried using a design involving 3 transmission gates per channel, and then one which had a 3-phase tree of transmission gates, but both of those have signals travel through 3 transmission gates, which in testing resulted in significant switching noise. Since we're operating in the domain of a maximum of around 10 kHz signals, after initial sweeps of the frequency and voltage ranges, it was determined that minimizing the size of the transistors still yielded acceptable levels of noise. As seen in Figure 19, having just a single phase of transmission gates leads to very minimal switching noise. Because of this, we went with the minimum channel length and twice the minimum width allowed by the PDK, so if need be, we could add dummy pass gates to the design.

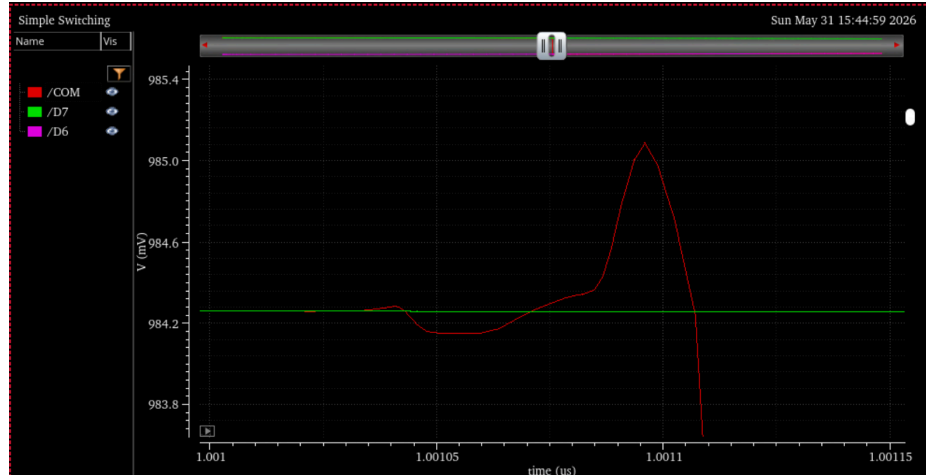


Figure 19: MUX Switching Noise

The physical layout of the MUX, as seen in Figure 20, routes the eight analog channels through the transmission gate array to a single common output line (COM) compactly.

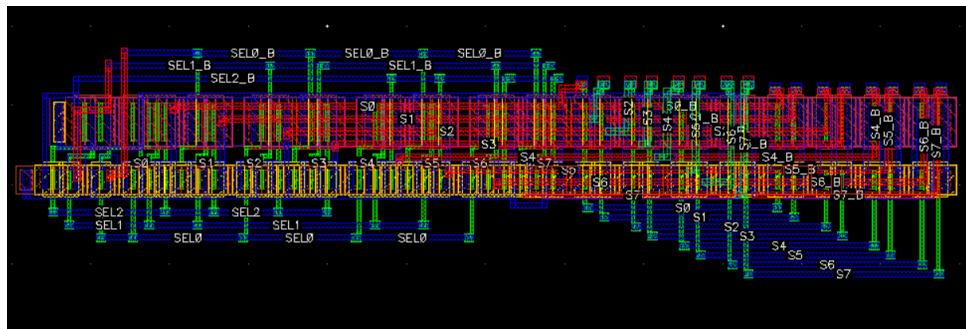


Figure 20: MUX Layout View

Analog-Digital Converter (ADC)

The ADC used in this design utilized a Successive Approximation Register (SAR) topology, allowing for smaller space consumption as compared to a Flash ADC. As mentioned, it was designed in a previous class (CE493) by a group member of this project. This ADC fits well because it was designed for low power, and can still run safely at frequencies far above the required sampling rate for this chip. The target application for the ADC is low-power biomedical and sensing systems, with design goals of 1 V supply, 0-1 V input range, 5 MS/s sampling rate, and ≤ 5 pJ per conversion step.

As a quick overview, the figure below shows the flow and operation. As an input voltage is sampled, it charges a capacitive DAC which feeds into a comparator, whose decision is part of a 8-cycle binary search. At the end of the 8 cycles, an 8-bit output register latches the correct 8-bit digital output. The schematic and layouts are provided in Figures 22 and 23, respectively.

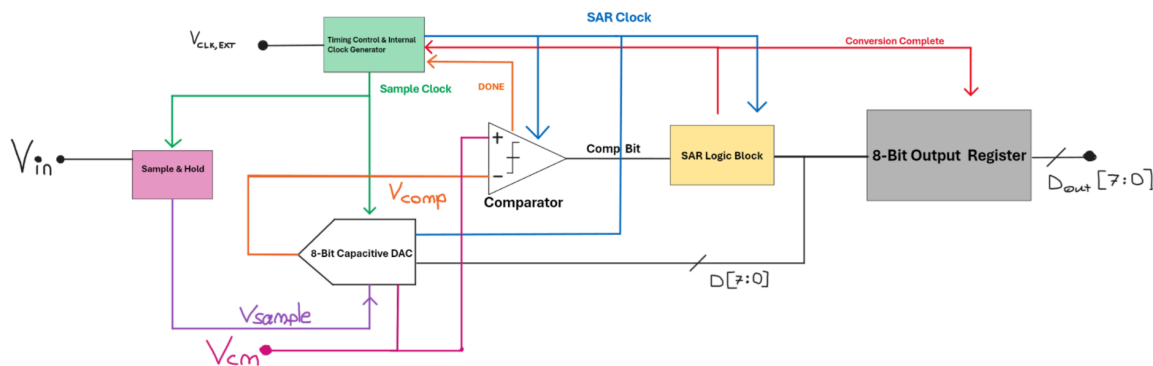


Figure 21: ADC Topology Overview

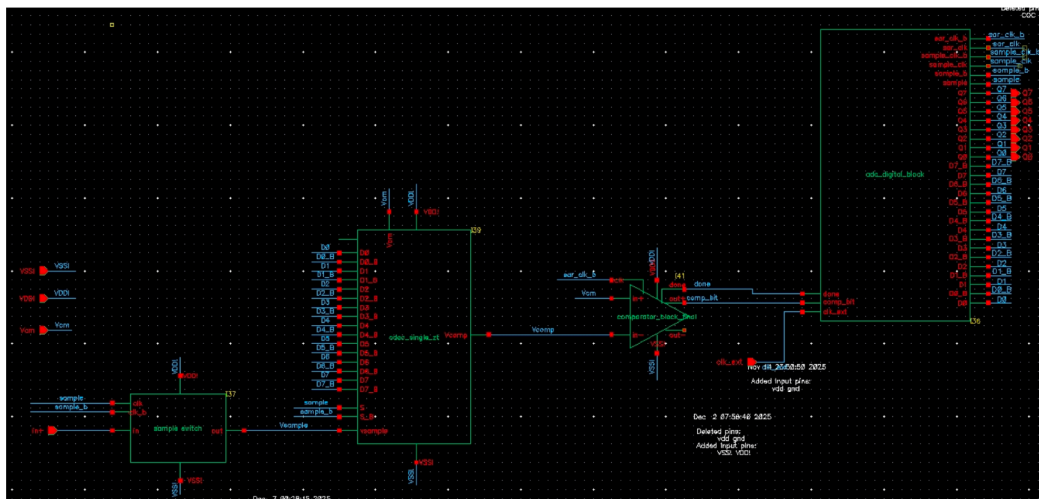


Figure 22: ADC Schematic in Virtuoso

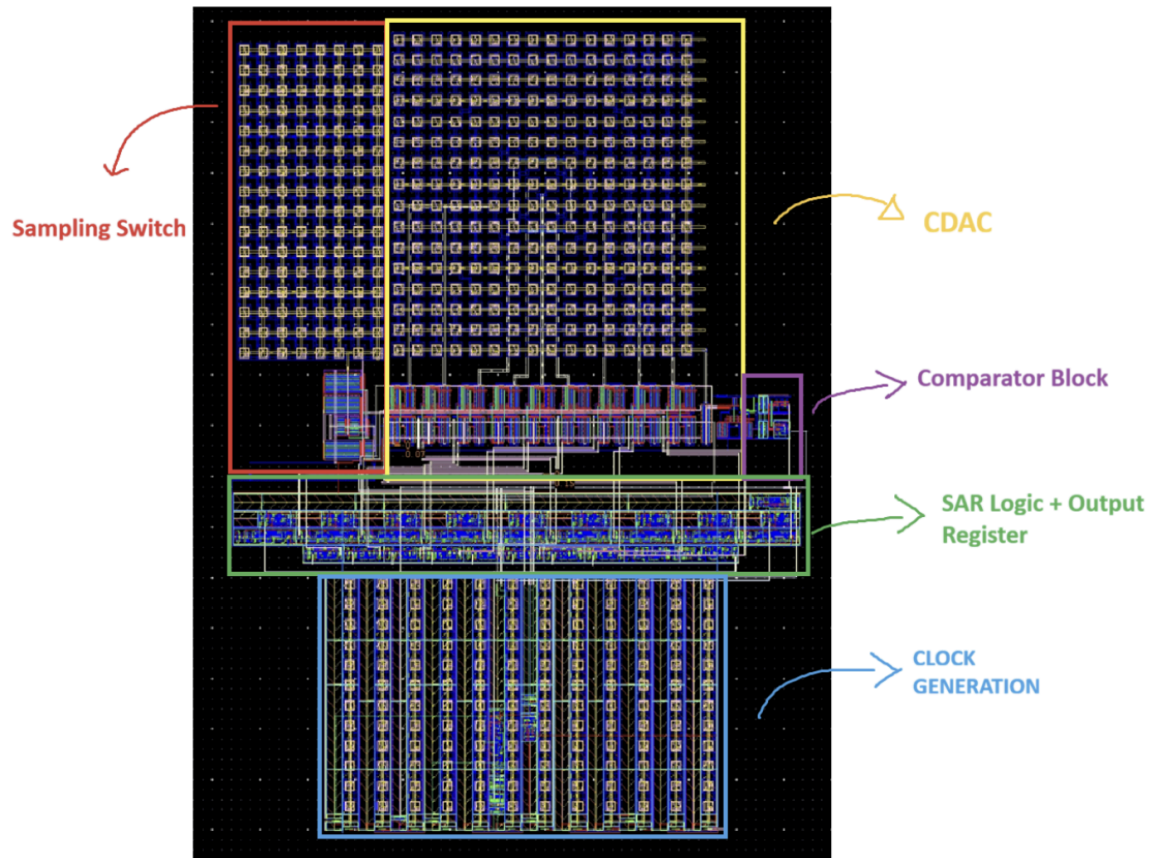


Figure 23: ADC Layout in Virtuoso

More in depth explanations, including verification for the ADC, can be found in Appendix A, along with a report.

Full Analog Mixed Signal Testing

The entire analog design was verified by a top level testbench that swept all frequency bands of interest and all amplitude ranges of interest. Figure 26 shows a condensed version of the results of the entire run. Figure 27 shows the initial input signals.

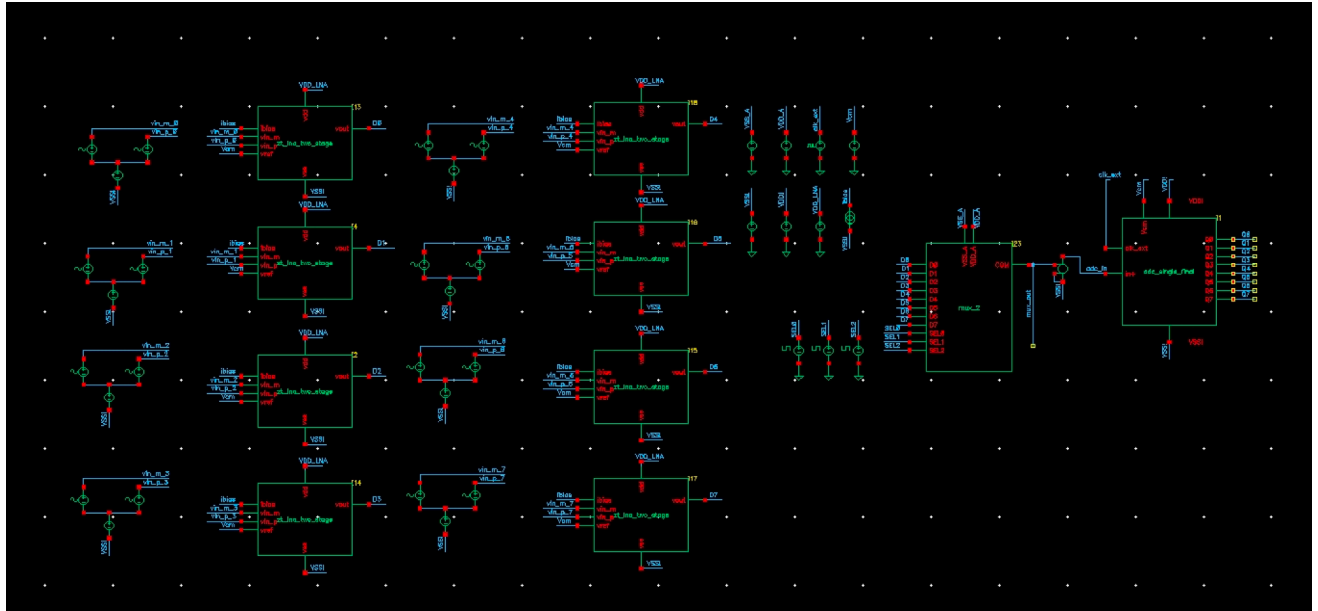
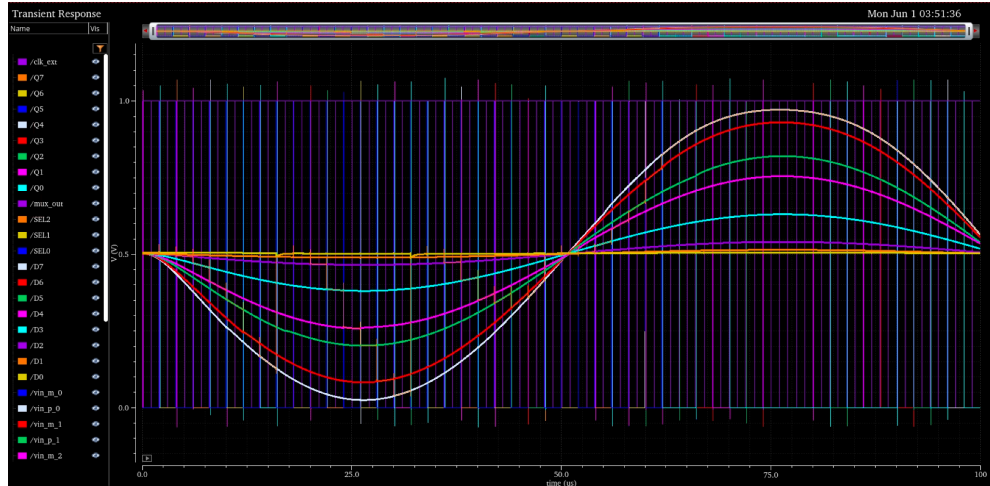


Figure 24: Analog Top Level Schematic

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
filter	filter	filter					
signal	signal	ack_ext					
signal	signal	/Q7					
signal	signal	/Q6					
signal	signal	/Q5					
signal	signal	/Q4					
signal	signal	/Q3					
signal	signal	/Q2					
signal	signal	/Q1					
signal	signal	/Q0					
signal	signal	/Q0_out					
signal	signal	/ack_in					
signal	signal	/SEL2					
signal	signal	/SEL1					
signal	signal	/SEL0					
signal	signal	/Q7					
signal	signal	/Q6					
signal	signal	/Q5					
signal	signal	/Q4					
signal	signal	/Q3					
signal	signal	/Q2					
signal	signal	/Q1					
signal	signal	/Q0					
signal	signal	/Q0_m_0					
signal	signal	/Q0_m_1					
signal	signal	/Q0_m_2					
signal	signal	/Q0_m_3					
signal	signal	/Q0_m_4					
signal	signal	/Q0_m_5					
signal	signal	/Q0_m_6					
signal	signal	/Q0_m_7					
signal	signal	/Q0_m_8					
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signal	signal	/Q0_m_11					
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signal	signal	/Q0_m_19					
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signal	signal	/Q0_m_21					
signal	signal	/Q0_m_22					
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signal	signal	/Q0_m_24					
signal	signal	/Q0_m_25					
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signal	signal	/Q0_m_33					
signal	signal	/Q0_m_34					
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signal	signal	/Q0_m_38					
signal	signal	/Q0_m_39					
signal	signal	/Q0_m_40					
signal	signal	/Q0_m_41					
signal	signal	/Q0_m_42					
signal	signal	/Q0_m_43					
signal	signal	/Q0_m_44					
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signal	signal	/Q0_m_46					
signal	signal	/Q0_m_47					
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signal	signal	/Q0_m_68					
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signal	signal	/Q0_m_72					
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signal	signal	/Q0_m_74					
signal	signal	/Q0_m_75					
signal	signal	/Q0_m_76					
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signal	signal	/Q0_m_92					
signal	signal	/Q0_m_93					
signal	signal	/Q0_m_94					
signal	signal	/Q0_m_95					
signal	signal	/Q0_m_96					
signal	signal	/Q0_m_97					
signal	signal	/Q0_m_98					
signal	signal	/Q0_m_99					
signal	signal	/Q0_m_100					

Figure 25: Analog Top Level Maestro Testbench Setup



In this testbench, the inputs are driven by off-phase inputs with varying amplitudes, ranging from 1 uVpp to 500 uVpp, covering the entire range of voltage inputs that a neural chip would intake.

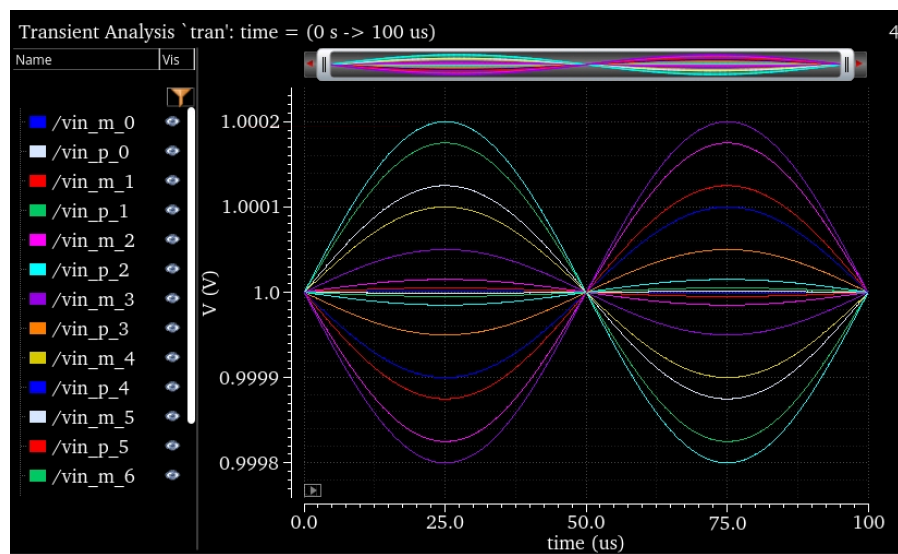


Figure 27: Input Voltages (Micro-volt range)

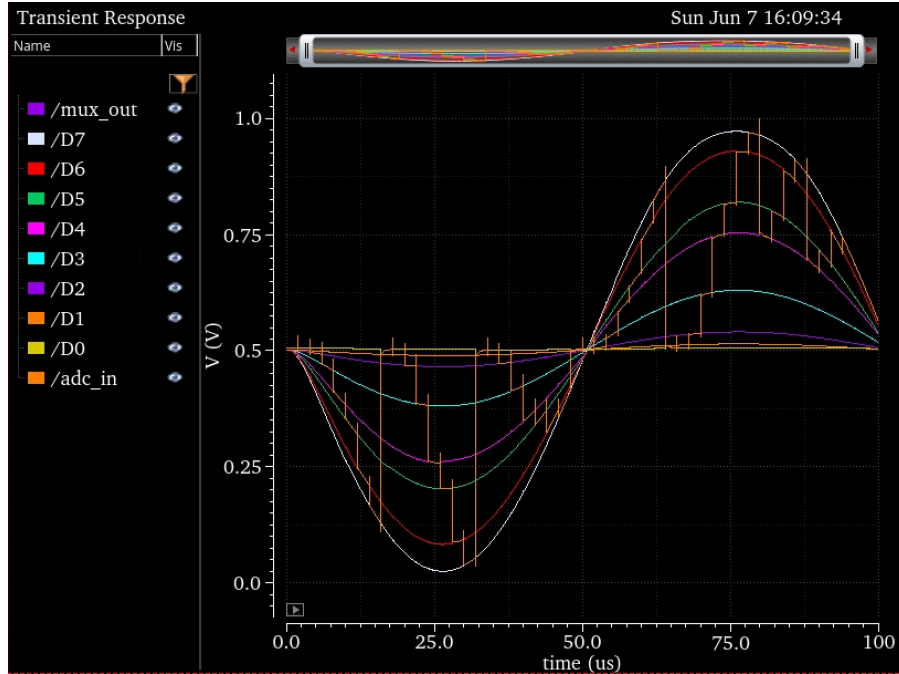


Figure 28: Output Voltages (Volt-range)

The ADC codes were verified by hand to match the input voltages, since Virtuoso has no smart tool to do so, and was tedious. However, once we matched up the input voltages to the output of the ADC, we were able to leverage a python script to calculate what that number corresponded to to verify that it reflected the correct output of the test vectors. Figure 29 shows the outputs of the ADC as Q0 to Q7 waveforms as digital signals. Figure 30 shows the MUX selection bits being toggled which result in the mux selecting a different LNA channel to output to the ADC, as noted by the jagged orange trace, which traces the mux switching.

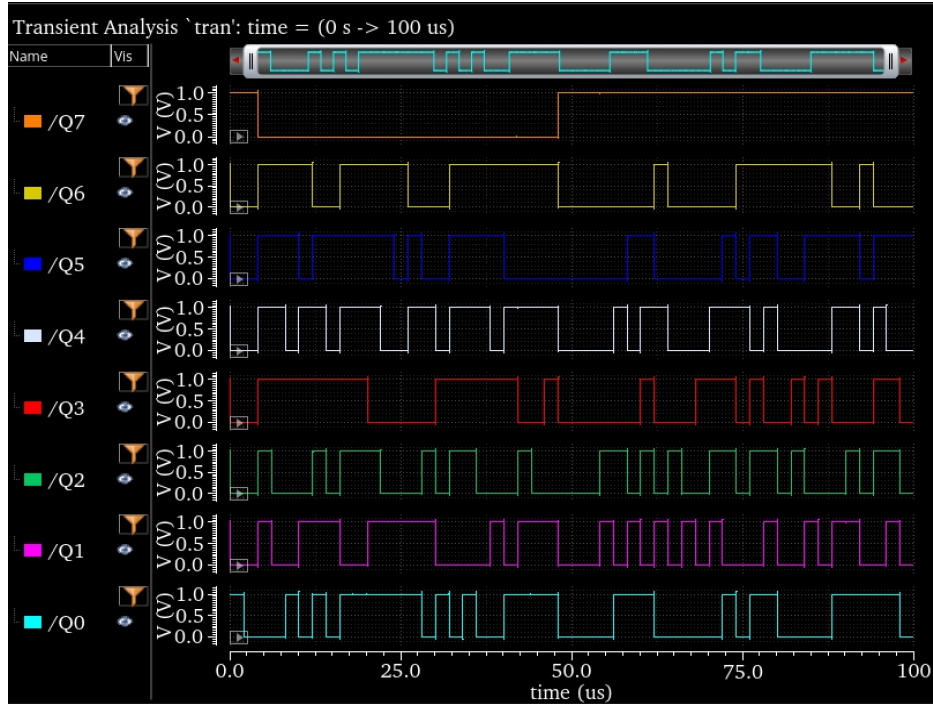


Figure 29: ADC Outputs

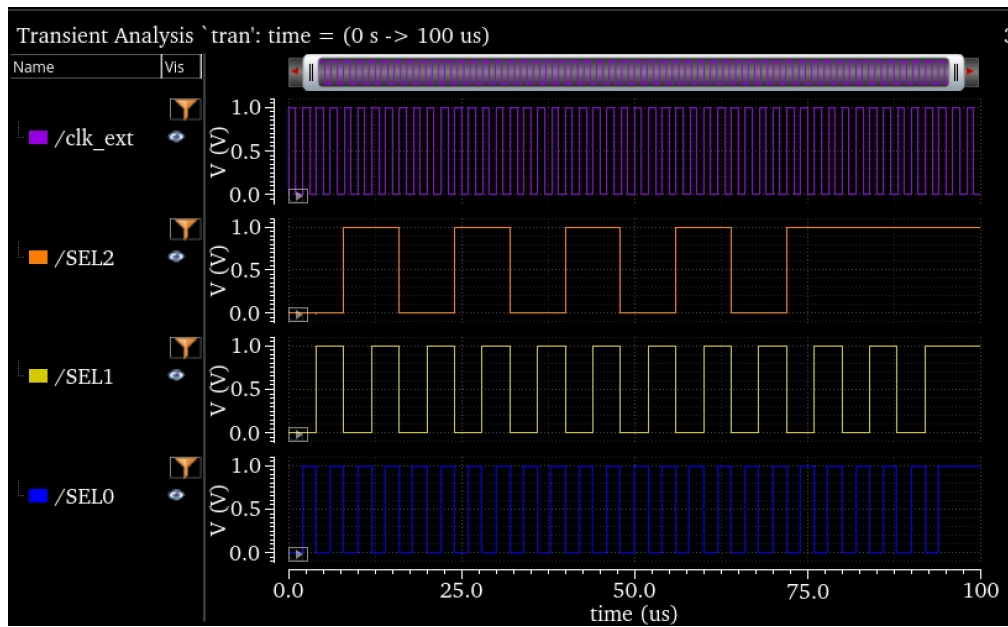


Figure 30: ADC Selection bits + Clock

Together, these results in the top-level testbench show that the analog front-end successfully does its job of amplifying a microvolt level signal to a 0-1V ADC, and the ADC produces a valid and accurate output. The output of the digital signals from the ADC feeds into the digital backend.

Digital Design

Digital Architecture Overview

The digital end is responsible for converting the digitized neural recordings into a real-time classification of motor intent. The system receives the 8-bit neural samples from the SAR ADC at the analog end, then processes them through a sequence of feature extraction, machine-learning inference, and communication stages.

The digital architecture has a pipeline design that includes five primary stages: sample collection, feature extraction, neural network inference, classification, and output transmission. The neural samples are collected into a fixed-length observation window before they are converted into a compact feature representation. Next, the features are processed by a quantized multilayer perceptron (MLP) classifier that produces a score corresponding to the different movement categories. Lastly, the decision logic selects the most likely classification, and the results are packaged and transmitted to any external controller using a SPI interface.

At the top level, the digital end integrates all the RTL modules into one classification pipeline. The top module coordinates the transfer of data between each processing stage and manages the control signals needed for operations. By connecting all the processing stages within a single RTL hierarchy, the top-level design provides a complete end-to-end path from neural signal acquisition to motor intent classification, while at the same time allowing each module to be independently designed, verified, and integrated. Figure 31 illustrates the RTL architecture of the digital end.



Figure 31: RTL Architecture

Machine Learning Model

As the chip's job is to decode motor intent from raw neural electrode signals in real time, we need to train a model that can perform the classification later on in the MLP stage. A classical lookup table or threshold rule can't generalize across electrode variation and trial-to-trial noise. A trained MLP, small enough to load entirely in a scan chain, and is proven to give robust, session-adaptive classification at negligible silicon cost.

Data Preparation and Preprocessing

The training process used neural recordings from the Brochier dataset, (Blackrock NS5 format, monkey-L), which contains 4 balanced grasp classes: Power Grasp (low/high force) and Spherical Grasp (low/high force). We decided to use this dataset to train because it was the only publicly available dataset that contained real voltages that was labelled, as we wanted to simulate as close as we could to a real end to end pipeline. We reduced the original 96-channel recording system to an 8-channel implementation that is suitable for a low-power brain-machine interface. To simulate the hardware to create testvectors, we used Python scripts to both process the data and train the model.

The first step of simulating the hardware was to isolate the spiking activity, where raw broadband neural signals were sampled first at 30 kHz and then processed using a bandpass filter from 300 Hz to 3 kHz. Next, the filtered signals were downsampled to 5 kHz to reduce computational complexity but also preserve the neural information needed for motor intent decoding. To simulate the on-chip ADC, the signals were clipped to the 1st and 99th percentiles and then quantized to 8-bit values ranging from 0 to 255. We extracted a 50 ms observation window from each recording trial, creating 250 samples per channel after downsampling. These processed samples form the input dataset used for feature selection and classifier training.

Feature Selection

A feature selection stage was used to identify the channels that provided the most useful information for movement classification because the original dataset had 96 recording channels. The method called Fisher Discriminant scoring was used to rank channels on their ability to separate the different movement classes. The Fisher score allows the channels with the strongest class-specific neural activity to be identified by measuring the ratio between inter-class separation and intra-class variance. The eight highest-ranked channels were chosen and then used throughout the machine learning phase and hardware design flow. These eight channels significantly lowered the hardware complexity, while still keeping the most relevant neural information for classification.

Network Training

To train the PyTorch model, we used a multilayer perceptron (MLP) classifier. PyTorch decodes the motor intent from the extracted neural features. The 8-neuron input layer network has a hidden layer containing 8 neurons with ReLU activation, and an output layer containing 4 neurons corresponding to the four movement classes.

We performed training by utilizing the Adam optimization algorithm with an 80/20 training and testing split. Adam was selected because it adaptively changes the learning rate of each network parameter,

improving convergence while requiring minimal hyperparameter tuning. The final classifier achieved about 70% classification accuracy on the test dataset.

Quantization and Hardware Deployment

The trained network weights were quantized to signed 8-bit integers following training to enable efficient implementation in digital hardware. Quantization helped decrease the memory requirements of the classifier while still keeping classification performance. This resulted in an estimated accuracy loss of about 3% compared to the floating-point model.

The Python training pipeline created the hardware deployment files used throughout the digital end. The trained new parameters were exported as weights.hex file for scan-chain loading into the chip, while test_vectors.txt files were created for RTL verification. Additionally, a cycle-accurate integer reference model was created to make sure that the quantized Python implementation produced the same results as the Verilog hardware implementation. Python script was also utilized to create per-subblock intermediate outputs for golden references.

Frontend Sample Collection

This is the start of the RTL portion of the project. The sample collection frontend acts as an interface between the analog and digital pipeline. As we share a single ADC, the samples must be reorganized into channel-specific data structures.

An output of this block is adc_channel, a 3-bit bus that drives the selection bits of the analog mux, allowing the digital side to control and know which channel it is sampling. Internally, the module maintains write counters and channel indexing logic to track the incoming samples and decide when the collection observation window has been completed. After all eight channels have accumulated 250 samples each, a window completion signal is asserted, indicating the feature extraction stage can now start processing the data. Instead of having a continuously streaming data source, this approach allows the downstream RTL blocks to operate on a fully assembled neural recording window, simplifying both feature extraction and neural network inference.

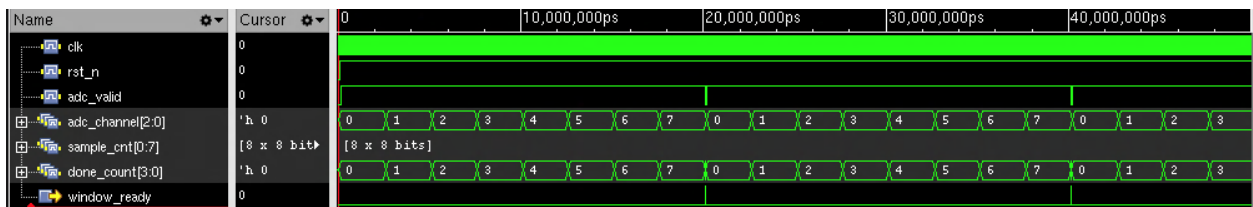


Figure 32: Sample Collection Waveform

Figure 32 shows the simulation waveform of the Sample Collection Frontend. The ADC samples are demultiplexed into eight channel buffers. The window_ready signal is asserted once the complete observation window is collected.

Spiking Band Power Feature Extraction

When a complete set of neural data is put together by the sample collection frontend, it gets sent to the Spiking Band Power (SBP) feature extraction stage. The purpose of the module is to reduce the complexity and unnecessary parts of the neural data while still keeping the information most relevant for motor intent classification. Instead of performing inference on all 2,000 samples directly, the SBP stage converts the raw neural recordings into a compact feature vector, allowing the machine learning classifier to process the data efficiently.

SBP is a broadband neural feature that estimates aggregate electrode spiking activity without requiring explicit spike detection or sorting. It is robust to low SNR conditions and electrode degradation, and is simple enough to implement in hardware without any multipliers. The SBP feature for each channel is defined as:

$$\text{SBP}[\text{ch}] = (\sum |x[n] - 128|) \gg 8$$

Here, $x[n]$ is the n -th ADC sample, 128 is the unsigned 8-bit midscale zero-signal reference, and the sum is taken over all $N = 250$ samples in the window. The right shift by 8 bits approximates division by 250 using a power-of-two normalization, avoiding any hardware divider. The hardware uses a single shared 16-bit accumulator that processes all 8 channels sequentially, one sample per clock cycle. The absolute deviation $|x[n] - 128|$ is computed combinationally by a subtract-and-mux: if $x[n] \geq 128$ the result is $x[n] - 128$, otherwise it is $128 - x[n]$. Thus, no multipliers are needed, and at the end of each channel's 250 samples, the upper 8 bits of the accumulator are captured as the SBP feature for that channel and the accumulator resets to zero for the next channel. The total latency is $8 \times 250 = 2,000$ clock cycles.

The resulting 8-element feature vector compresses the full 2,000-sample ADC window into 8 scalar values representing the mean neural activity level per channel, which are passed directly to the MLP classifier.

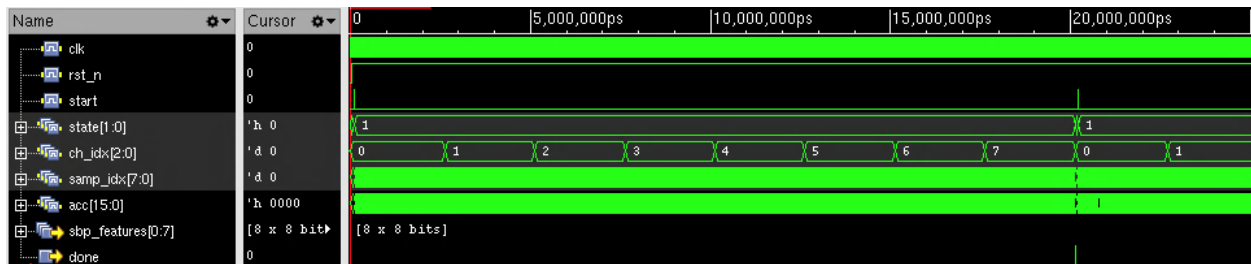


Figure 33: Feature Extraction Waveform

Figure 33 shows the simulation waveform of the SBP Feature Extraction. Each channel's neural samples are processed to generate the eight-dimensional feature vector used by the ML classifier.

Multilayer Preceptron Inference Engine

After the SBP feature extraction stage creates the eight-dimensional feature vector, the data is then passed to the multilayer perceptron (MLP) inference engine. This module's role is to classify the neural activity

patterns and figure out which of the four grasp categories used through this project. Rather than relying on simple thresholding or linear classification, the MLP captures more complex relationships between the intended movement and the extracted neural features. The ML model is the one trained in the section above.

The MLP consists of two fully connected layers implemented entirely in RTL. The first layer receives the eight SBP feature values as inputs and maps them to eight hidden neurons. Each of these hidden neurons calculates a weighted sum of the input features and applies the ReLU activation function. In this hidden layer, the network is able to capture the nonlinear relationships within the neural data, which improves the classification performance. The second layer then takes these hidden activations and creates four output scores that correspond to the four different movement classes: PG-LF, PG-HF, SG-LF, and SG-HF. Each score represents the relative confidence of the classifier for each type of movement. These scores are then utilized to determine the final prediction.

To make the hardware work efficiently, the network uses quantized 8-bit signed integers to represent all of the network weights and activations. These trained weights were generated during the Python model development stage and are loaded into the chip using a scan-chain interface during initialization. Storing the weights separately from the RTL implementation allows the classifier to be easily reconfigured for different neural decoding tasks without having to modify the underlying hardware architecture.

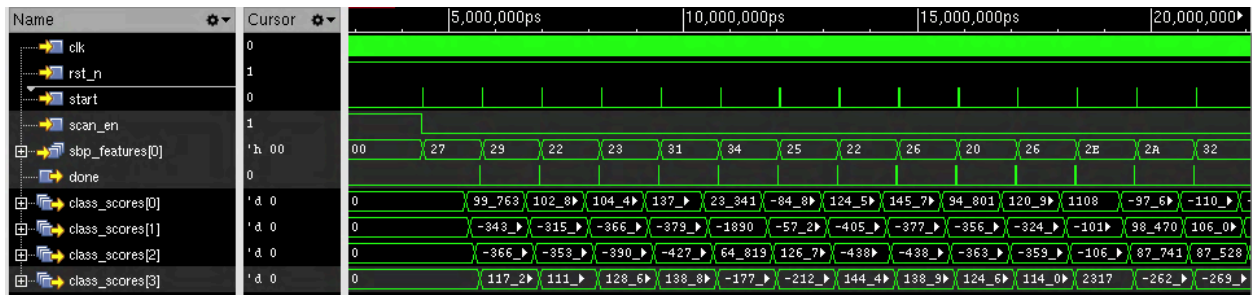


Figure 34: MLP Inference Waveform

Figure 34 shows the simulation waveform of the MLP Inference Engine. The quantized feature inputs are processed in the hidden and output layers to produce classification scores for the four movement categories.

Output Layer and Argmax Decision Logic

After the MLP inference engine generates the four output scores, these values are passed to the Argmax decision module, whose role is to determine which movement class has the highest predicted confidence and convert the neural network output into the final classification result. The Argmax module compares the four scores, which correspond to the four movement classes, from the MLP engine and determines the largest value. The class identified with the highest score is then selected as the predicted motor intent.

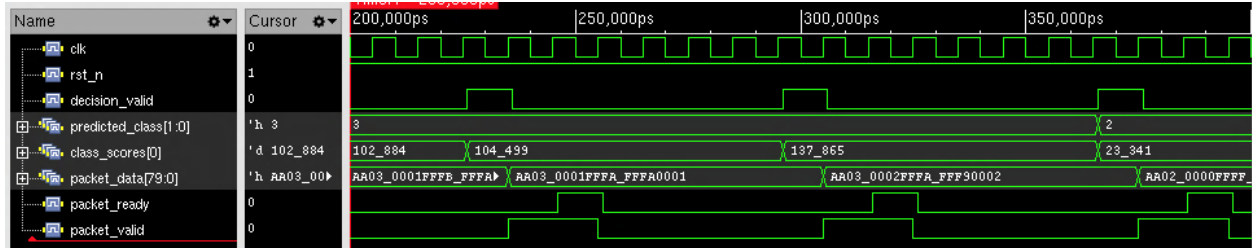


Figure 35: Output Formatter Waveform

Figure 35 shows the simulation waveform of the Output Formatter module. Both the neural network output scores and classification results are assembled into a fixed 10-byte packet to be transmitted.

Output Formatter and SPI Interface

Once the predicted movement class is determined from the Argmax module, the classification result is passed to the Output Formatter and SPI Interface. This stage packages the classification information into structured data format and transmits the results to the external control for further processing.

The Output Formatter takes the classification results and organizes them into a 10-byte packet containing the following: synchronization header, predicted movement class, and confidence scores generated by the neural network. By including both the final classification and the output scores, we get more information that's used for debugging, system verification, and downstream processing. Before transmitting the data, the formatter assembles all this information into a predefined packet structure.

To communicate with external devices, a Serial Peripheral Interface (SPI) is used due to its simple implementation and its ability to provide reliable high-speed data transfer. In order for the decoded motor intent to be used for prosthetic control, external computation, and other downstream applications, the SPI interface serializes the formatted packet and transmits the data to an external processor.

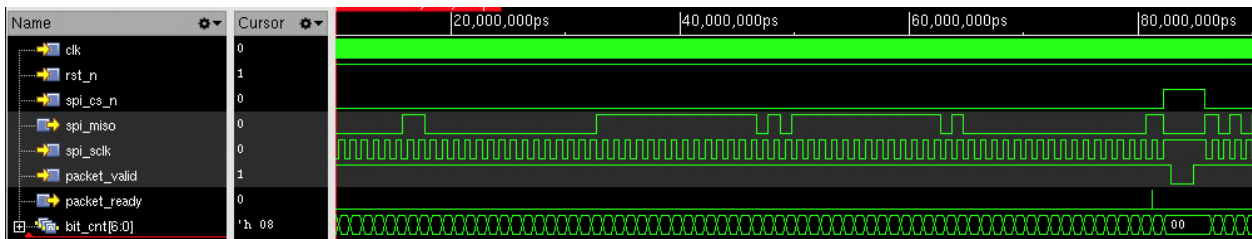


Figure 36: SPI Interface Waveform

Figure 36 shows the simulation waveform of the SPI interface. The formatted classification packet is serialized and transmitted using the SPI communication protocol.

RTL Verification

Before integrating the complete classification pipeline, RTL verification was performed to ensure each digital block operated correctly using SystemVerilog testbenches in Cadence Xcelium along with the expected outputs generated by the Python reference model. We leverage a python script that produces

intermediate golden vectors so that we can debug the RTL block by block instead of having to debug the entire thing from the top level.

First, we performed verification at each of the module levels. Each module has its own corresponding individual testbench and was supplied with representative input stimuli, and the resulting outputs were compared against expected values generated from the software models and hand calculations. This process ensured that each block functioned correctly before integrating them into one top module.

After completing module-level verification, the full digital end was simulated using end-to-end test vectors created from the machine learning pipeline. These test vectors went through the entire classification process, from sample collection to SPI packet generation. We then compared the resulting outputs against the Python reference implementation to ensure functional correctness of the complete RTL design. All module-level and top-level simulations produced outputs that matched the expected Python reference model results, demonstrating functional correctness of the RTL implementation.

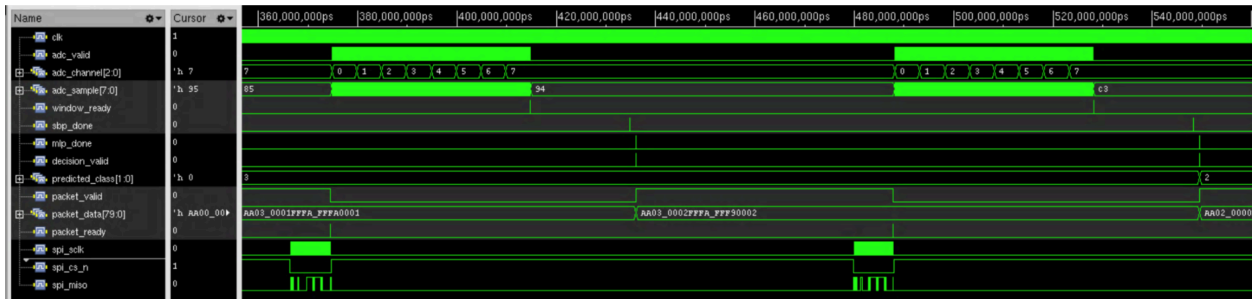


Figure 37: Digital Block Waveform

Figure 37 shows the top-level RTL simulation waveform of the full integrated digital backend. Neural samples progress through the complete classification pipeline from acquisition to packet generation and SPI transmission.

Digital Backend Implementation

For the digital backend section, we used Cadence Genus for synthesis, and Cadence Innovus for Place-and-Route (PNR).

Timing Constraints

Timing constraints were using a Synopsys Design Constraints (SDC) file. We designed the digital backend to operate at a target frequency of 50 MHz, with a clock period of 20 ns. We chose this operating frequency to comfortably satisfy the throughput requirements of the neural decoding pipeline while still leaving enough timing for synthesis and place-and-route. The details of why we chose the 50 MHz target can be found in the Constraints section.

To ensure proper functioning, the main clock system was supplemented with separate clocks for the SPI interface and scan-chain programming logic. To prevent false timing paths between asynchronous clock domains and ensure accurate timing analysis during synthesis, we applied appropriate timing constraints. These constraints allowed the synthesis tools to maintain reliable operation across all functional blocks and optimize the design. The timing constraints established the performance targets used throughout the backend design flow, including logic synthesis, gate-level verification, and physical implementation. The digital backend could be optimized for both timing closure and correction by defining these requirements early in the design process.

Logic Synthesis

After applying the timing constraints, we used Cadence Genus to synthesize the RTL design with the GPDK45 cell library. This synthesis process converts the behavioral RTL description into a gate-level netlist made up of standard cells, all while optimizing for timing, area, and power. This step allowed the digital backend to be evaluated in terms of operating performance and hardware resources before the physical implementation.

Total Standard Cells	41,354 cells	sample_collection	20,391 cells 156,675 μm^2 58.6%
Total Cell Area	207,681.9 μm^2	sbp_feature_extrac	16,582 cells 66,698 μm^2 25.0%
Total Netlist Area	267,194.4 μm^2 (about 517 $\mu\text{m} \times$ 517 μm)	MLP + argmax + fmt + SPI	4,381 cells 43,821 μm^2 16.4%

Figure 38: Area Utilization

Figure 38 (left) shows the area utilization of the synthesized digital backend. The Sample Collection Frontend occupies the largest portion of the design area due to storage of the complete neural observation window.

Figure 38 (right) illustrates the area utilization of the synthesized design. The digital backend consists of 41,354 standard cells and occupies a total netlist area of 267,194.4 μm^2 . As expected, the Sample Collection Frontend accounted for the largest portion of the design, covering 58.6% of the total area because the module is responsible for storing and managing the complete neural observation window, which comprises 2,000 samples. The SBP Feature Extraction block accounted for 25% of the total area, while the remaining 16.4% is shared among the MLP inference engine, Argmax logic, Output Formatter, and SPI Interface.

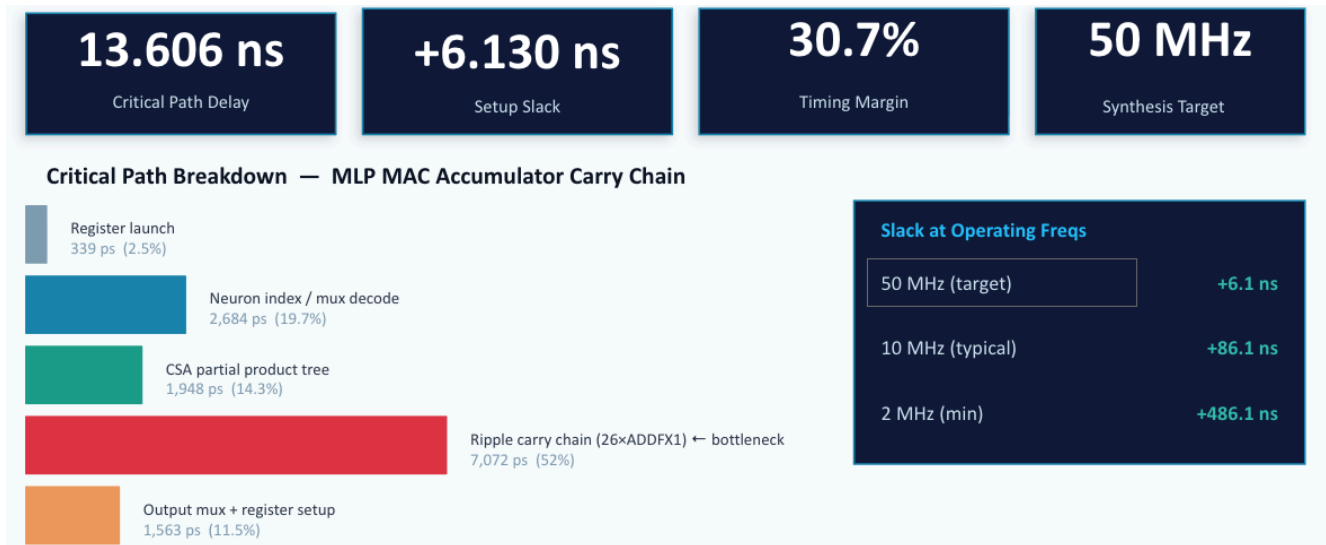


Figure 39: Timing Analysis

Figure 39 shows the timing analysis results following synthesis. The digital backend achieves positive setup slack at the target operating frequency, demonstrating successful timing closure.

Timing analysis results are shown in Figure 39. The synthesized design met the target operating frequency of 50 MHz with a positive setup slack of 6.130 ns and achieved a critical path delay of 13.606 ns. The timing margin of 30.7% indicates that the design satisfies the required timing constraints. The longest delay occurred within the MLP multiple-accumulate data path, more specifically within the accumulator carry chain used in the neural network inference.

These successful synthesis results prove that the complete digital classification pipeline can be implemented efficiently in hardware while meeting the performance requirements necessary for real-time neural decoding.

Physical Implementation

After the logic synthesis, the generated gate-level netlist was imported to Cadence Innovus for physical implementation. The physical design flow consisted of the following: floorplanning, power distribution network generation, standard cell placement, clock routing, and signal routing. To preserve the timing and functional requirements established during synthesis, these steps transformed the synthesized netlist into a physically realizable layout.

First, a floorplan was created to define the core area and placement region for the digital backend. Using dedicated routing resources, power and ground connections were distributed through the design to ensure reliable operation of all the standard cells. After placement, we performed routing to establish the connectivity between the synthesized logic cells and complete the physical implementation of the classification pipeline.

After routing, we performed timing and connectivity checks to verify the integrity of the design. The resulting layout met the timing requirements established during synthesis. The layout also provided a physically realizable implementation of the backend, allowing for integration with the analog front-end.

Latency Analysis

Stage	Cycles	Time	Relative Duration
Sample Collection 8ch × 250 samples	2,000	200 μs	
SBP Extraction 8ch × 250 samples + 1	2,001	200 μs	
MLP Inference 8→8→4 time-muxed MAC	~110	11 μs	
Argmax 1-cycle comparison tree	1	0.1 μs	
Output Formatter 10-byte SPI packet	1	0.1 μs	
SPI Transmission 80 bits @ SCLK	~810	81 μs	
TOTAL per classification	~4,923	~492 μs	→ ~2,032 classifications / second

Figure 40: Latency Analysis

Figure 40 shows the latency contribution of each stage in the digital classification pipeline. Sample collection and feature extraction account for the majority of the overall classification latency.

To determine the overall throughput of the system, we analyzed the latency of each stage in the digital classification pipeline. Figure 40 shows how much classification time each processing block used. The whole pipeline requires about 4,923 clock cycles, with a total classification latency of about 492 μs.

The Sample Collection Frontend and SBP Feature Extraction account for most of the overall latency, each requiring about 200 μs to process all the data in one observation window. This behavior is expected because both modules operate on all 2,000 neural samples collected in each classification cycle. On the other hand, the MLP inference engine only requires about 11 μs to generate its prediction, while Argmax and Output Formatter stages can complete their operations in a single clock cycle. The SPI transmission requires about 81 μs to transmit the complete 10-byte classification packet, performing the final stage of the pipeline.

Overall, the digital backend produces a classification rate of about 2,032 classifications per second. The results demonstrate that the implemented architecture is able to support real-time neural decoding while still maintaining a low processing latency.

Final Layout

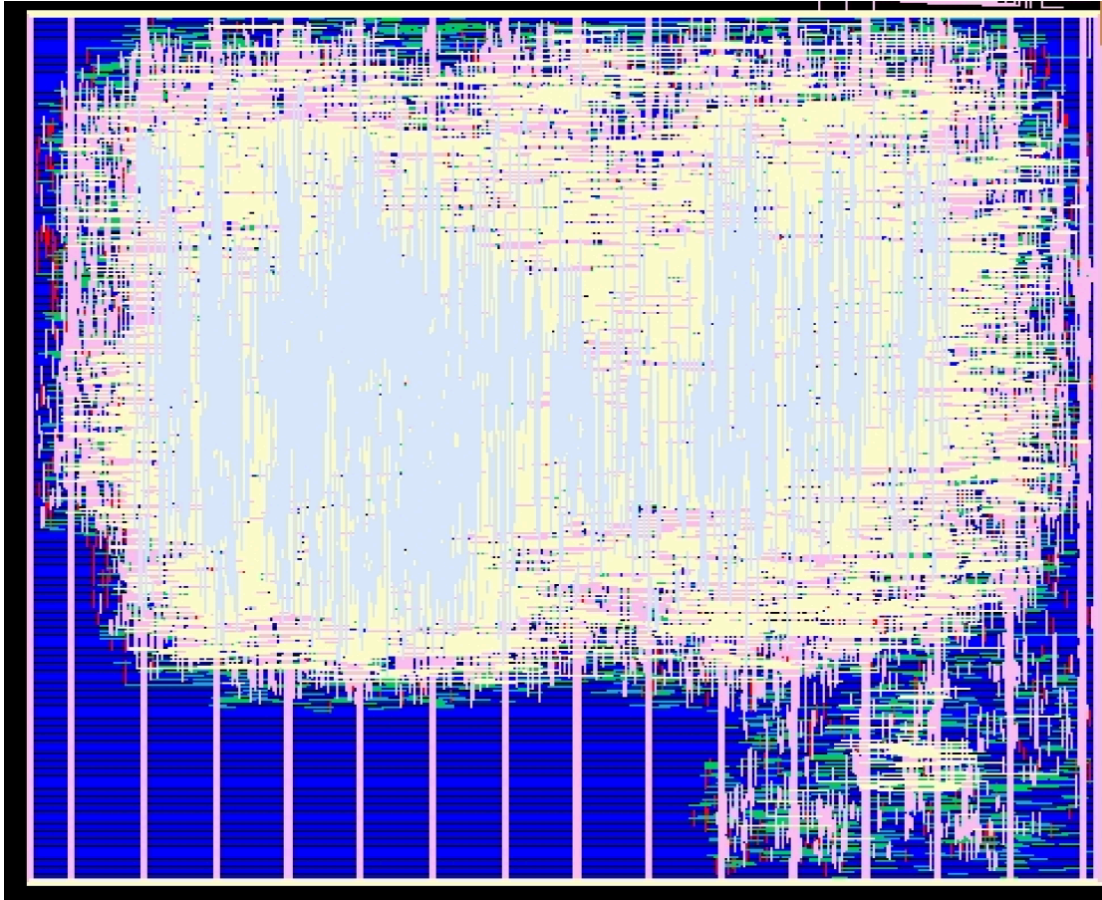


Figure 41: Final Layout

Figure 41 shows the final routed layout of the digital backend after place-and-route in Cadence Innovus using the GPDK45 standard-cell library.

In Figure 41, the completed digital layout after place-and-route is shown. After the logic synthesis and physical implementation in Cadence Innovus, all of the standard cells were successfully placed and routed while still satisfying the timing constraints established during the synthesis stage. The resulting layout represents the final physical realization of the digital classification pipeline, including the following modules: the Sample Collection Frontend, SBP Feature Extraction, MLP Inference Engine, Argmax Decision Logic, Output Formatter, and SPI interface.

The final routed design occupies a total netlist area of $267,194.4 \mu\text{m}^2$ and maintains a positive timing slack at the target operating frequency of 50 MHz. The successful routing of the design proves that all the modules can be physically integrated while still preserving the functionality verified during RTL simulation. The final implementation confirms that the complete neural decoding pipeline can be realized efficiently in hardware using the 45nm standard cell library.

The finalized digital backend was then integrated with the analog front end to create the complete mixed-signal brain-machine interface system. The analog and digital subsystems together provide an end-to-end solution capable of acquiring neural signals, extracting their features, performing machine learning inference, and transmitting classification results all in real time.

Power Analysis

Post-route power analysis was performed in Cadence Innovus using the final placed-and-routed digital backend layout. The analysis used extracted post-route parasitics and the slow timing library at a detected digital supply voltage of 0.9 V. Innovus performed a vectorless activity-propagation estimate. Therefore, the reported values should be interpreted as first-order power estimates rather than final activity-based signoff power. Due to the time constraints of this course, we were unable to perform dynamic power analysis. The following is produced from static power analysis.

Total Power						
Total Internal Power:		4.00938900	73.6736%			
Total Switching Power:		1.42734970	26.2279%			
Total Leakage Power:		0.00535808	0.0985%			
Total Power:		5.44209671				
Group		Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential		3.037	0.1558	0.003012	3.196	58.73
Macro		0	0	0	0	0
IO		0	0	0	0	0
Combinational		0.2235	0.5677	0.001867	0.7931	14.57
Clock (Combinational)		0.7487	0.7039	0.0004781	1.453	26.7
Clock (Sequential)		0	0	0	0	0
Total		4.009	1.427	0.005358	5.442	100
Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	0.9	4.009	1.427	0.005358	5.442	100
Clock		Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
scan_clk		0.004978	0.01164	6.614e-06	0.01662	0.3054
clk		0.7437	0.6922	0.0004715	1.436	26.39
spi_sclk		0	0	0	0	0
Total		0.7487	0.7039	0.0004781	1.453	26.7
* Power Distribution Summary:						
* Highest Average Power:		FE_OFC5495_scan_clk (CLKBUF20):			0.00137	
* Highest Leakage Power:		u_sbp/g1214295_5703 (NOR4X8):			7.22e-07	
* Total Cap:		2.71524e-10 F				
* Total instances in design:		46179				
* Total instances in design with no power:		0				
* Total instances in design with no activity:		0				
* Total Fillers and Decap:		0				
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00, mem(process/total)=1740.91MB/1740.91MB)						

Figure 42: Static Power Analysis

The Innovus power report estimated a total digital backend power of 5.442 mW under the constrained clock setup. This total consists of 4.009 mW internal cell power, 1.427 mW switching power, and 5.36

μW leakage power. Thus, the power is dominated by dynamic switching activity, while leakage contributes less than 0.1% of the total power. The estimated average current at 0.90 V is:

$$I_{avg} = P_{total} / VDD = 5.442 \text{ mW} / 0.90 \text{ V} = 6.05 \text{ mA}$$

Because the design was constrained with a 50 MHz main clock, this 5.442 mW value represents a conservative high-frequency estimate. The intended BMI operating clock can be lower, so dynamic power can be approximately scaled with frequency. Dynamic power is the sum of internal and switching power:

$$P_{dynamic} = 4.0009 \text{ mW} + 1.427 \text{ mW} = 5.437 \text{ mW}$$

Leakage power is approximately frequency-independent, so the total power at lower clock rates is estimated as:

$$P_{total}(f) \approx P_{leakage} + P_{dynamic} \times (f / 50 \text{ MHz})$$

At 10 MHz, the estimated total power is:

$$P_{total}(10M) = 0.00536 \text{ mW} + 5.437 \text{ mW} \times 0.20 = 1.093 \text{ mW}$$

At 2 MHz, the estimated total power is:

$$P_{total}(2M) = 0.00536 \text{ mW} + 5.437 \text{ mW} \times 0.04 = 0.223 \text{ mW}$$

The final digital backend footprint is approximately $596.6 \mu\text{m} \times 596.6 \mu\text{m}$, corresponding to an area of 0.00356 cm^2 . Using this area, the estimated digital power density is:

- At 50 MHz: $5.442 \text{ mW} / 0.00356 \text{ cm}^2 = 1529 \text{ mW/cm}^2$
- At 10 MHz: $1.093 \text{ mW} / 0.00356 \text{ cm}^2 = 307 \text{ mW/cm}^2$
- At 2 MHz: $0.223 \text{ mW} / 0.00356 \text{ cm}^2 = 62.6 \text{ mW/cm}^2$

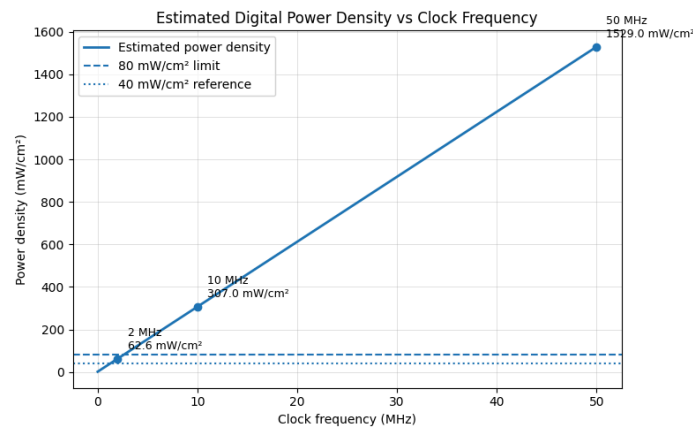


Figure 43: Power Density Ceiling vs Clock Frequency

The graph in Figure 43 was created in Python to plot the digital power density vs clock frequency. This power-density comparison is important because implanted neural electronics are constrained by local tissue heating. Literature on brain implants commonly discusses safety limits around 40-80 mW/cm² or related temperature-rise constraints to avoid tissue damage from heating. Under the vectorless estimate, 50 MHz and 10 MHz operation exceed an 80 mW/cm² local power-density target when considering only the digital macro area, while 2 MHz operation falls below this limit. Reducing the operating clock to the lowest rate that still supports the ADC/MUX sampling, feature extraction, MLP inference, and SPI output significantly reduces dynamic power and improves thermal safety. A future refinement would be to run activity-based power analysis using VCD/SAIF switching activity from realistic BMI input vectors, which would provide a more accurate estimate than vectorless activity propagation.

Whole Chip Integration (AMS + Digital)

Whole chip integration was accomplished by importing the digital backend from Innovus into a Virtuoso library via Stream. Then, the top level schematic and layout was completed. Passing both DRC and LVS ensured that the chips had the correct connectivity. The top level schematics and layouts are shown in Figure 44 and respectively.

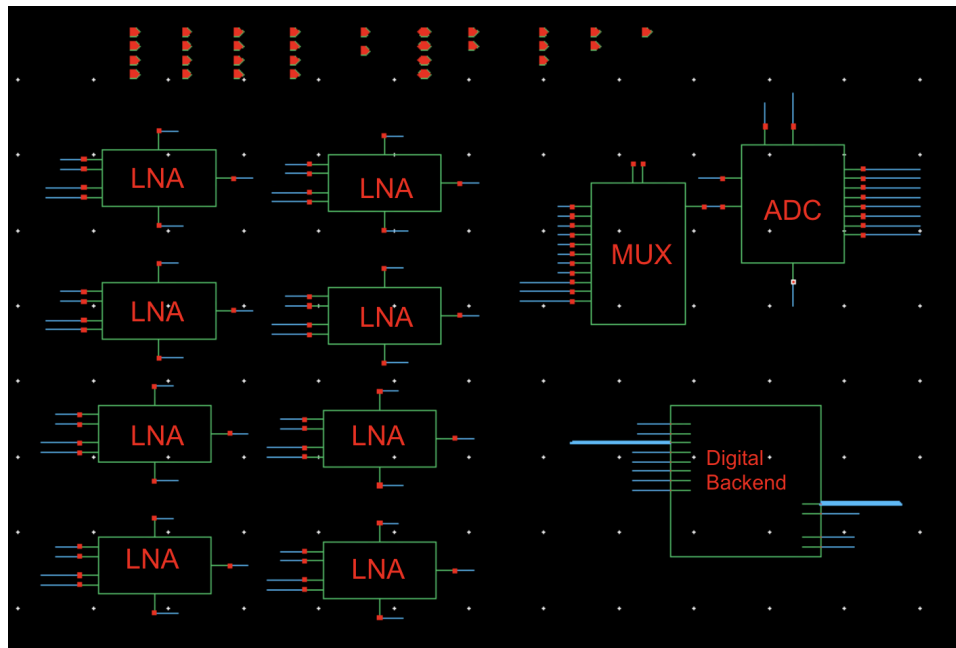


Figure 44: Top Level Chip Schematic

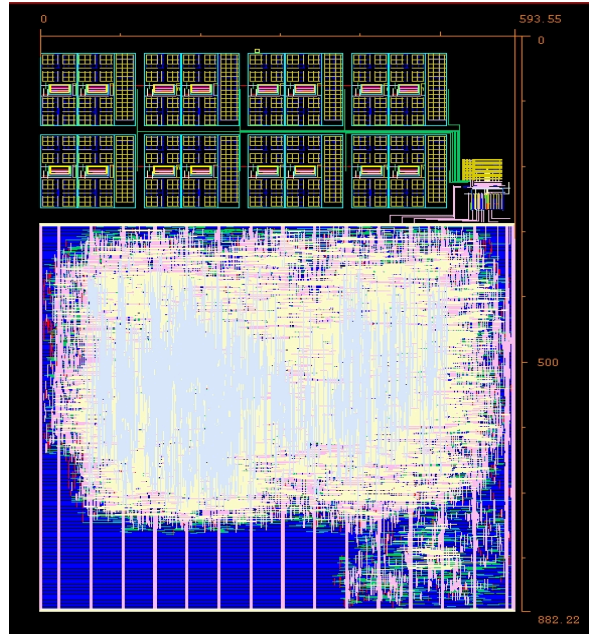


Figure 45: Top Level Chip Layout

The design considerations were mainly on the layout side, as the schematic level changes were already synthesized and designed in their respective sections. After completing the layout, we ran design rules check (DRC) and layout-versus-schematic (LVS) and parasitic extraction (PEX). We were unable to test the results of the fully integrated analog and digital design together, but we were able to ensure that the LVS, DRC, and PEX were well extracted, and that they were able to pass their individual simulations on their separate selves.

Design Optimizations

The analog front end did not meet its gain specification on the first design pass. A single capacitively-coupled stage produced only about 37 dB of gain, short of the 40 dB targeted per stage, and far short of the total gain needed to bring a microvolt-scale neural signal up to the ADC's 1 V_{pp} full-scale range. Fixing that issue, as well as several others, led to the design optimizations below:

Cascading Two LNA Stages

The first finding was that one LNA stage simply could not reach the required gain: at roughly 37 dB ($\sim 70\times$) per stage, a single amplifier left the neural signal well below the ADC's input range. Rather than push a single stage harder, which would have cost headroom, bandwidth, and stability, the per-channel amplifier was modified to cascade two identical LNA stages. The cascaded chain reaches approximately 65 dB (over $1500\times$) post-parasitic-extraction, enough to span the ADC's full-scale range, while reusing the same stage twice kept the layout regular and tileable across the eight channels.

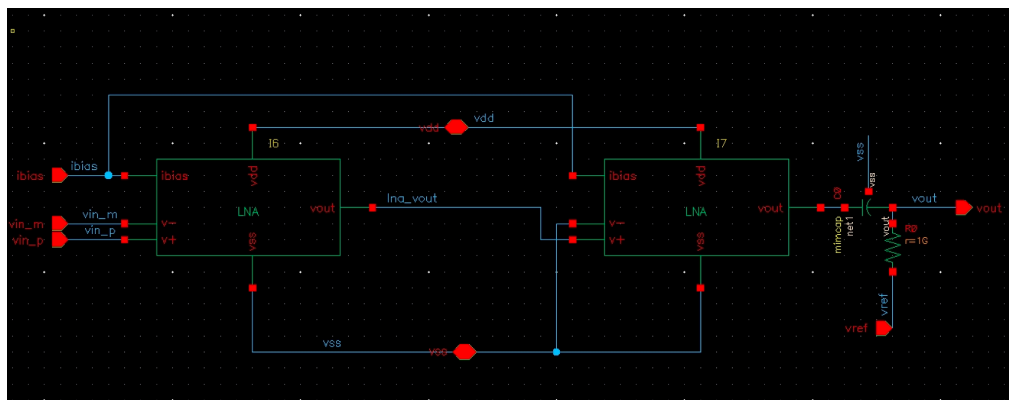


Figure 46: LNA Cascading to Achieve Sufficient Gain

Inter-Stage High Pass Network

Cascading two high-gain stages created a second problem: any DC offset or operating-point drift at the first stage's output is amplified by the second stage, pushing the signal off-center and wasting ADC range. To correct this, a high-pass filter built from MOS-bipolar pseudoresistors was added to pull the DC level of the node toward the common-mode voltage V_{cm} while still passing the neural band. This re-centers the signal on the midpoint of the ADC's input range so the full 1 V_{pp} swing is usable. This was an architectural choice, not simply a fix because the pulldown fixes the output common mode at V_{cm} , the amplifier shares a single bias reference with the ADC instead of generating its own, simplifying biasing and guaranteeing the amplifier output sits exactly where the ADC expects its input.

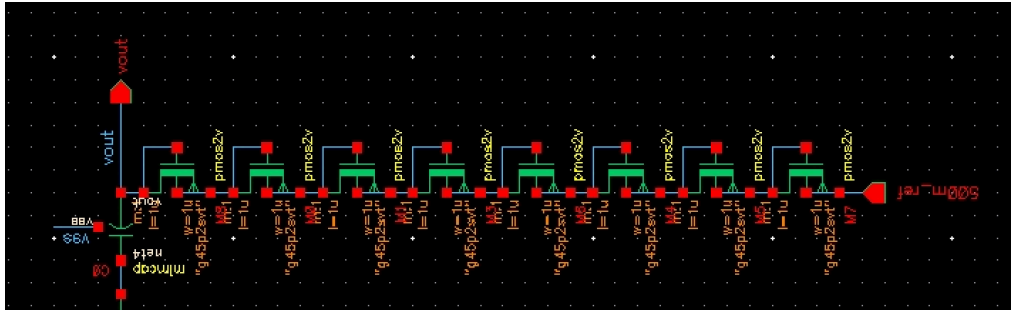


Figure 47: Pseudoresistor Pulldown and Output Capacitor

Bandwidth Adjustment

Our initial design simulations saw severe bandwidth issues, with the first pole of the system response being too high of a frequency where we saw somewhat significant signal attenuation at the bottom of our desired frequency band. This first pole is the result of the very high value high pass input filter on the LNA input, set by the AC coupling capacitance and pseudoresistance to ground. To resolve this issue, we had to shift the frequency of that first pole, which was adjusted by adding another pseudoresistor in series to increase the total series resistance by 33%, decreasing the pole location by about 25%, entirely resolving the issue.

Engineering Standards

The development of the Learned Feature BMI ASIC spans many engineering standards, including digital design, analog design, mixed signal integration, and device safety. To ensure reliability and adherence to the industry's best practices, this project implemented and was guided by several standards.

Digital Design and Hardware Description Languages

IEEE 1800-2017

This standard dictates the syntax and semantics for System Verilog. This standard was used extensively to write RTL code in the various blocks of the digital backend, including sample collection, SBP extraction, MLP inference, Argmax calculation, and SPI communication. Furthermore, the standard was used when making testbenches in Xcellium to test each block's functional correctness.

IEEE 1364-2005

The netlists made in Genus for each block adhere to this standard. This ensured that the synthesized gate level representations of the RTL code could be accurately simulated and mapped to the GPDK45 standard cells.

Physical Design and ASIC Synthesis Standards

Graphic Design System (GDS) Format

This is the industry standard database design format and the end result of the Innovus place and route of the digital block. This format was used to transfer the digital block design into Virtuoso for mixed signal integration, verification and testing.

Library/Design Exchange Format (LEF/DEF)

These formats detail the rules and contractions of the standard cells of the process design kit. We used these in the place and route portions of our designs in Virtuoso layout.

Synopsys Design Constraints (SDC)

This is the standard that defines timing, power, and area constraints for logic synthesis. Custom SDC files were made to dictate our three clock domains: the 50 MHz system clock, the 10 MHz SPI clock, and the 25 MHz scan chain clock. Additionally, it outlined the IO delays the synthesis tool needed to optimize data paths in the digital designs.

Communication Protocols and Testability

Serial Peripheral Interface (SPI)

We used SPI to transmit the 10 byte neural classification as the final output for our entire mixed signal block. The design strictly adheres to the SPI standards, using a 10 MHz clock, a chip select, and a master in slave out line for reliable, low overhead data offloading.

IEEE 1149.1

Commonly known as Joint Test Action Group (JTAG). We used the principles of this standard to make our design more testable. We implemented a 25 MHz serial scan chain to load the 108 byte quantized MLP weights into the chip's internal flip flops, using the shift register method mentioned in this standard.

Safety Standards

Because the end result of this project is to have an ASIC for use in medical applications, we constrained the device in accordance with medical safety standards during the initial brainstorming phase and throughout the design of the device.

IEC 60601-1

This standard concerns the electrical and thermal safety of medical devices. Most importantly, the average power density of our design determines whether the device complies with the maximum safe heat flux of 80 mW/cm² needed by tissue heating limits, as dictated by this standard. By optimizing our digital logic and scaling our clock frequency to 2 MHz, our maximum digital power density ended up being 62.6 mW/cm², satisfying thermal standards for safe patient use.

ISO 14971

This standard details the identification and mitigation of operational risks. In our layout, we mitigate cross domain interference by strictly separating the analog and digital power domains. This prevents digital switching noise from corrupting sensitive neural signals, which would otherwise result in unreliable intent classifications by the device.

ISO 10993

The ASIC's LNA was explicitly designed to interface with the Blackrock Neurotech Neuroport Electrode Array, which is fully implanted inside a patient. This standard lays out the biocompatibility that these invasive electrodes were designed with, and the hermetic packaging that would eventually be required to house a device using this chip. We followed this standard as far as the scope of the project allowed. Future work would require extensive adherence to this standard due to the nature of the electrodes.

Ethical Standards

Because this technology interfaces directly with the human motor cortex to decode intended actions, it raises concerns regarding data privacy, algorithmic bias and patient safety. Our device is more of a prototype than something that can directly be used in medical applications, so a clinical deployment of this ASIC would require strict safeguards to mitigate risks.

Data Privacy and Security Vulnerabilities

Intracortical neural data is highly sensitive, dealing with a patient's thoughts and in this case, the final output is the user's physical intentions. If malicious actors were to intercept or manipulate this data, the consequences could range from privacy violations to the hijacking of the devices this ASIC feeds data to. In its current state, our ASIC transmits the classification packet via an unencrypted SPI. Furthermore, the MLP weights are loaded via a scan chain that lacks authentication.

To prevent data theft and manipulation in a clinical setting, future versions of this chip must incorporate an on chip cryptographic hardware accelerator (such as AES-256) to encrypt the SPI output payload before it leaves the digital backend. Additionally, a secure boot mechanism and cryptographic authentication protocol must be implemented for the scan chain to ensure that only verified weights can be loaded into the device, preventing malicious reprogramming.

Algorithmic Bias

The 8 bit quantized MLP block on this chip was trained on a highly specific dataset: parallel recordings from a macaque motor cortex during an instructed delayed reach to grasp task. A major ethical concern is algorithmic bias; a model trained on a limited, non-human dataset will not inherently generalize safely to diverse human populations with varying neurological profiles and/or physical injuries.

To mitigate bias and ensure equitable efficacy, clinical translation would require extensive revalidation and retraining of the model using diverse human datasets spanning different demographics, ages, and causes of paralysis. Because our architecture stores its weights in a reconfigurable scan chain rather than hardwiring them, the hardware supports improvement towards a safer, less biased algorithm, allowing clinicians to continually upload personalized models tailored to individual patients.

Physical Safety

Unlike a standard electronic device, a BMI directly controls physical actuators such as a robotic arm or wheelchair in the patient's environment. A false positive classification could cause a neuroprosthesis to crush an object or injure the user or a bystander.

Training and verification testing must be conducted to minimize the false positive rate. At the system level, mitigation requires utilizing the raw output scores from the MLP inference block. Rather than blindly trusting the Argmax decision, the external controller must enforce a confidence threshold. If the margin between the top two predicted classes is too narrow, the system should default to a safe state rather

than executing an uncertain, potentially dangerous movement. Furthermore, physical force limiters must be implemented on the downstream hardware to prevent harm in the event of an unavoidable classification error.

Accessibility

BMI technology carries the risk of widening socioeconomic health disparities. If the technology is prohibitively expensive, the restorative benefits of real time neural decoding are restricted to only those who can afford it, leaving behind many people who also suffer from paralysis and other physical disabilities.

While surgical and clinical costs remain a hurdle, our hardware design directly addresses the financial accessibility of the device itself. By consolidating the analog frontend, feature extraction, and ML classification onto a single 0.524 mm² die, we help mitigate the cost of a final prosthesis solution.

Conclusion

This report presented the design of a fully integrated mixed-signal brain-machine interface ASIC implemented in a 45 nm GPDK CMOS process. The chip addresses a fundamental gap in the existing BMI hardware literature by integrating a custom analog neural recording frontend with an on-chip nonlinear classifier on a single device, eliminating the dependence on external analog front-end chips that dominates system power in prior work such as NeuRAD.

The final design successfully met and exceeded its original specifications. The analog frontend achieved around 65 dB of gain, isolating the 300-10,000 Hz neural band while adjusting for the DC offset of the electrode interface. In the digital domain, the classification pipeline was fully synthesized, placed and routed, complying with GPDK45 DRC and LVS rules. The logic cleared 50 MHz with 6.13 ns of slack. By scaling the operating frequency of the design to 2 MHz, which supports the 200 kHz ADC sampling requirement, the digital block's power density drops to 62.6 mW/cm², safely below thermal dissipation limits for medical use.

This ASIC serves as a robust proof of concept, but many avenues exist for future improvement, like more complex mixed signal testing, or using more specific data to train and better our algorithm to increase accuracy and decrease bias.

Design Improvements

Due to the physical limitations in our team and the time span we were able to work on the design, the current design would benefit from future improvements. We noted down several of these improvements that can be realized with more time spent or investment into the project for future iterations.

1. Monte Carlo Simulations

Simulations across process, voltage, and temperature (PVT) corners would quantify the statistical spread of gain, bandwidth, input-referred noise, and ADC linearity, providing yield estimates and confirming design margin before fabrication.

2. Low Impedance Buffer for MUX /ADC

The LNA output currently drives the SAR ADC sampling capacitor directly through the MUX, creating a kickback path that can disturb the LNA output node during the sample-and-hold phase. Inserting a dedicated low-impedance buffer between the MUX output and ADC input would isolate the LNA from capacitive DAC switching transients and reduce reliance on ideal bias current and voltage reference assumptions, improving ADC settling accuracy and overall signal chain robustness.

3. Full System-Level Noise Simulations

We were unable to complete a full system-level input-referred noise analysis integrating contributions from the pseudoresistors, OTA, MUX switch resistance, and ADC quantization noise. This analysis is necessary to confirm that the end-to-end noise floor remains below the typical extracellular neural background noise of 5-10 μ V_{rms}, and should be accompanied by a complete power breakdown across all analog blocks to validate implantable power budget compliance.

4. Real intracordial signals as test cases

The digital backend was verified using vectors from the given Brochier dataset, and is postprocessed in Python rather than real recorded neural signals passed through the analog frontend. Acquiring real intracortical recordings and constructing Verilog-A behavioral models of the LNA, MUX, and ADC would allow end-to-end simulation with realistic signal statistics, capturing nonlinearities and noise that are not represented in the current testbench.

5. Full Analog-digital simulations

The analog and digital domains were verified independently, the analog frontend in Cadence Virtuoso and the digital backend in Xcelium. A unified analog-mixed-signal (AMS) simulation using a tool such as Cadence Spectre AMS would co-simulate both domains simultaneously, catching interface-level issues such as ADC metastability, MUX settling time violations, and clock domain interactions that are invisible when each domain is verified in isolation.

Future Work

Looking ahead, the most impactful feature extension would be expanding the classifier beyond four output classes to support finer motor control granularity. This chip was architecturally designed with this expansion in mind, as different weights could be scanned in for different applications. Decoding individual finger movements, wrist rotations, or combined hand-arm gestures would significantly increase the clinical utility of the device for dexterous neuroprosthetic applications. This would require retraining the MLP on a richer labeled dataset and widening the output layer, both of which are straightforward given the reconfigurable scan-chain weight loading mechanism already implemented in the current design.

Despite these limitations, this chip demonstrates that a complete neural signal acquisition and classification pipeline, from electrode interface to SPI output, can be realized on a single mixed-signal ASIC at power and area levels consistent with the trajectory toward fully implantable brain-machine interfaces. The integration of a fully custom analog frontend with a reconfigurable on-chip classifier represents a meaningful step toward a self-contained, low-power neural decoding device suitable for chronic implantation.

References

- [1] R. R. Harrison and C. Charles, "A low-power low-noise CMOS amplifier for neural recording applications," IEEE Journal of Solid-State Circuits, vol. 38, no. 6, pp. 958-965, June 2003.
- [2] J. Kim et al., "A 16-Channel Fully Configurable Neural SoC with 1.52 $\mu\text{W}/\text{Ch}$ Signal Acquisition, 2.79 $\mu\text{W}/\text{Ch}$ Real-time Spike Classifier, and 1.79 TOPS/W Deep Neural Network Accelerator in 22nm FDSOI," IEEE Transactions on Biomedical Circuits and Systems, vol. 16, no. 1, pp. 94-107, Feb. 2022.
- [3] H. An et al., "A Power-Efficient Brain-Machine Interface System with a Sub-mW Feature Extraction and Decoding ASIC Demonstrated in Nonhuman Primates," IEEE Transactions on Biomedical Circuits and Systems, vol. 16, no. 3, pp. 317-330, May 2022.
- [4] M. Bulfer et al., "A 192-Channel 1D CNN-Based Neural Feature Extractor in 65nm CMOS for Brain-Machine Interfaces," Center for Sensing to Intelligence (S2I), California Institute of Technology, Feb. 2025.

Appendix

Appendix A: ADC Design Files

ADC design files can be found at the following Google Drive Link:

<https://drive.google.com/file/d/1g4mbYhf0YLxcuesPHs6LFbwZ1daA23Ea/view?usp=sharing>

Appendix B: Digital Design Files

All design files for the digital portion are located in the following path:
/home/vcs5888/CE392/bmi_chip/digital

RTL Source — /home/vcs5888/CE392/bmi_chip/digital/rtl/

File	Description
bmi_chip_top.sv	Top-level RTL integration
sample_collection.sv	Sample collection module
sbp_feature_extractor.sv	SBP feature extraction module
mlp_inference.sv	MLP inference engine
argmax.sv	Argmax classifier
output_formatter.sv	Output packet formatter
spi_slave.sv	SPI slave interface

Simulation — /home/vcs5888/CE392/bmi_chip/digital/sim/

File	Description
tb_bmi_chip_top.sv	Top-level testbench
run_rtl.sh	RTL simulation run script
run_gls.sh	Gate-level simulation run script
weights.hex	MLP weight file

Synthesis — /home/vcs5888/CE392/bmi_chip/digital/syn/

File	Description
bmi_chip_top_syn.v	Post-synthesis netlist
area.rpt	Area report
timing.rpt	Timing report
bmi_chip_top.sdc	Timing constraints

Backend / Place & Route — /home/vcs5888/CE392/bmi_chip/digital/pnr

File	Description
bmi_final_layout_with_pins.enc.dat	Final routed layout database
bmi_chip_top_final.gds	Final GDS layout
bmi_with_pins_clean.drc.rpt	DRC report

Appendix C: Digital Backend Design Summary

BMI Chip — Results Summary

Technology: GPDK045 / GSCLib045

Standard Cell Library Tools: Cadence Genus 18.14, Xcelium 18.09, Innovus 16.20

Synthesis (Genus)

Metric	Value
Critical path delay	6.771 ns
Slack vs 10 ns target	+3.124 ns
Total area	178,170 μm^2
Resettable flops	~16,741
Un-reset flops	~864 (mostly scan chain)
Critical path location	MLP MAC accumulator carry chain

Area by Block:

Block	Area (μm^2)
sample_collection	~134,664
sbp_feature_extraction	~26,466
MLP + argmax + formatter + SPI	~16,040
Total	178,170

Gate-Level Simulation (Xcelium)

Metric	Value
Test vectors passed	40 of 40
Class predictions	Correct, varied across 0/1/2/3
Mode	Zero-delay functional

Place and Route (Innovus)

Metric	Value
Core utilization	63%
Standard-cell instances	36,319 logic + filler cells
DRC violations	0
Post-route setup WNS	Met (positive slack)
Hold WNS	~-0.130 ns (minor violations, see note)

Overall

- Full synthesis, GLS, and place and route completed successfully with GPDK045/GSCLib045.
- GLS passes 40 of 40 test vectors with correct class predictions across all four output classes.
- 0 DRC violations in the final layout.
- Setup timing meets 50 MHz with positive slack.
- Final layout saved to bmi_final_layout_fixed.enc.dat; netlist (bmi_chip_top_nophy.v) and SDF (bmi_chip_top.sdf) generated.

_.1 Environment Setup

```
source /vol/ece303/genus_tutorial/cadence.env
```

```
cd ~/bmi_chip
```

_.2 Logic Synthesis (Cadence Genus)

```
read_hdl -sv ../sample_collection.sv
```

```
read_hdl -sv ../sbp_feature_extractor.sv
```

```
read_hdl -sv ../mlp_inference.sv
```

```
read_hdl -sv ../argmax.sv
```

```
read_hdl -sv ../output_formatter.sv
```

```
read_hdl -sv ../spi_slave.sv
```

```
read_hdl -sv ../bmi_chip_top.sv
```

```
set_db library \  
  
/vol/eecs391/GPDK045/gsclib045_all_v4.8/gsclib045/timing/slow_vdd1v0_basicCells.lib  
  
set_db lef_library \  
  
/vol/eecs391/GPDK045/gsclib045_all_v4.8/gsclib045/lef/gsclib045_macro.lef  
  
elaborate  
  
current_design bmi_chip_top  
  
read_sdc ../bmi_chip_top.sdc  
  
syn_generic  
  
syn_map  
  
syn_opt  
  
report_timing > timing.rpt  
  
report_area > area.rpt  
  
write_hdl > bmi_chip_top_syn.v
```

_.3 Physical Implementation (Cadence Innovus)

```
set init_gnd_net VSS  
  
set init_pwr_net VDD  
  
set init_verilog \  
  
/home/riw5160/bmi_chip/Synthesis_GPDK/bmi_chip_top_syn.v  
  
set init_top_cell bmi_chip_top  
  
init_design  
  
floorPlan -r 1.0 0.63 2 2 2 2  
  
globalNetConnect VDD -type pgpin -pin VDD -inst *  
  
globalNetConnect VSS -type pgpin -pin VSS -inst *  
  
addRing -nets {VSS VDD} \
```

```
-type core_rings \  
-follow io  
  
placeDesign  
  
optDesign -preCTS  
  
clockDesign \  
  
-specFile /vol/ece303/encounter_tutorial/Clock.ctstch  
  
optDesign -postCTS  
  
routeDesign -globalDetail  
  
verify_drc  
  
verifyConnectivity  
  
saveDesign bmi_final_layout_fixed.enc
```

_.4 Gate-Level Simulation

```
xrun -64bit \  
-access r \  
  
-xmelab_args "-warnmax 0 -delay_mode zero -maxdelays" \  
  
backend_GPDK/bmi_chip_top_nophy.v \  
  
tb_bmi_chip_top.sv \  
  
/vol/eecs391/GPDK045/gsclib045_all_v4.8/gsclib045/verilog/slow_vdd1v0_basicCells.v
```

_.5 Reopening Final Layout

```
restoreDesign \  
  
/home/riw5160/bmi_chip/backend_GPDK/bmi_final_layout_fixed.enc.dat \  
  
bmi_chip_top
```